

Investigating the potential risk of cadmium exposure on Osteoporosis: An integrated multi-omics approach

Yiwei Li^a, Xuezhen Liang^a, Yifa Rong^a, Kai Jiang^a, Jiahao Zhang^a, Gang Li^{b,*}

^a The First Clinical Medical School, Shandong University of Traditional Chinese Medicine, Jinan, Shandong, China

^b Orthopaedic, Affiliated Hospital of Shandong University of Traditional Chinese Medicine, Jinan, Shandong, China

ARTICLE INFO

Edited by Yong Liang

Keywords:

Cadmium
Osteoporosis
Network toxicology
Single-cell RNA sequencing
Geniposide

ABSTRACT

Osteoporosis (OP) is a chronic progressive bone disease, and its occurrence and development under cadmium exposure remain unclear. This study aims to explore the role of cadmium exposure in the pathogenesis of OP through a comprehensive analysis of multi-omics data. Through cross-sectional analysis using National Health and Nutrition Examination Survey (NHANES), we observed that cadmium exposure is a risk factor for OP. Bioinformatics and machine learning further emphasized the importance of FOXO3, CCND1, MAP1LC3B, HMOX1, and MT1G as independent risk factors for OP, which was confirmed through robust internal validation. Single-cell RNA sequencing revealed the heterogeneous expression of cadmium-related genes in different cell populations, with a particular emphasis on the role of HMOX1 in cell communication and signaling. Through Gene Set Enrichment Analysis (GSEA), we found that HMOX1 is positively correlated with the activity of M2 macrophage polarization. Through Mendelian randomization (MR), molecular docking, and molecular dynamics simulations, we further discovered that geniposide can target and bind to HMOX1. Ultimately, this comprehensive study elucidates the role of cadmium exposure in OP and highlights the potential of HMOX1 as a therapeutic target in alleviating the adverse effects of the disease.

1. Introduction

Cadmium is a toxic heavy metal that is widely present in the environment and can pose a serious threat to human health (Genchi et al., 2020). With the acceleration of industrialization, cadmium is not only extensively released into the environment through industrial activities, but is also widely present in consumer goods. Epidemiological studies suggest that food and cigarettes are the two main routes of cadmium intake (Zhu et al., 2024). Cadmium is associated with an increased risk of various diseases and can significantly raise the risk of chronic conditions such as cardiovascular diseases, kidney damage, and cancer (Satarug et al., 2010; Verzelloni et al., 2024). Cadmium has a long half-life, allowing it to accumulate in human organs over time and is considered a key inducer of inflammation, oxidative stress, and other processes (Ma et al., 2022).

Osteoporosis (OP) is a common skeletal disease characterized by decreased bone mass and deterioration of bone microstructure, leading to increased bone fragility and a significantly higher risk of fractures. Recent research has focused on the impact of cadmium exposure on bone

metabolism. A metabolomics study has shown that cadmium exposure can block the electron transport chain in human osteoblast-like cells, resulting in pyruvate accumulation and the buildup of nucleic acids associated with aging (Tian et al., 2021). Additionally, cadmium exposure activates the DNA damage response through phosphorylation of ATM and H2AX, while promoting the acetylation of SOD2, which induces mitochondrial dysfunction and exacerbates the aging process of osteoblasts (Zhou et al., 2023). Further research has revealed the critical role of NF-κB in the senescence of bone marrow-derived mesenchymal stromal cells induced by cadmium exposure (Luo et al., 2021). A systematic review and meta analysis found that even low levels of environmental cadmium exposure (urinary cadmium >0.5 µg/g creatinine) are a risk factor for OP (Kunioka et al., 2022). A case-control study found that blood cadmium levels are associated with an increased risk of fractures (Wallin et al., 2024). A study based on the Swedish cohort of the Osteoporotic Fractures in Men Study (MrOS) revealed that cadmium exposure is associated with reduced cortical thickness, cortical area, and trabecular bone volume fraction in elderly men (Wallin et al., 2021). A similar phenomenon has been validated in the Korean population (Kim

* Corresponding author.

E-mail addresses: liyiweizy@163.com (Y. Li), 60170109@sducm.edu.cn (X. Liang), 2018128134@sducm.edu.cn (Y. Rong), 2023111093@sducm.edu.cn (K. Jiang), zjh160037@foxmail.com (J. Zhang), sdszylg@163.com (G. Li).

<https://doi.org/10.1016/j.ecoenv.2025.118502>

Received 27 February 2025; Received in revised form 20 May 2025; Accepted 9 June 2025

Available online 14 June 2025

0147-6513/© 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

et al., 2021). Another predictive model found that cadmium exposure plays an important role in predicting decreased bone mineral density (BMD) (Ximenez et al., 2021). Human biomonitoring surveys emphasize that cadmium exposure represents a significant proportion of the societal costs associated with OP (Ougier et al., 2021). As an environmental pollutant, cadmium may play a crucial role in the onset and progression of OP.

This study employed various research methods aimed at exploring the potential role of cadmium exposure in OP. Through network toxicology analysis, we constructed a gene-toxicity association network to gain deeper insights into the expression patterns of cadmium exposure-related genes in OP and their potential toxicological mechanisms. Key genes were identified through machine learning, and bioinformatics and single-cell RNA sequencing were combined to reveal the complex relationship between cadmium exposure and mechanisms related to OP, such as M2 macrophage polarization. Single-cell sequencing analysis further revealed the specific expression of cadmium exposure-related genes across different cell types in OP, thereby enhancing understanding of the underlying pathological mechanisms. In addition, Drug target Mendelian randomization (MR) analysis, molecular docking, and molecular dynamics simulations were employed to provide clues for the development of novel targeted therapeutic drugs in the future.

2. Method

2.1. Cross-sectional analysis

National Health and Nutrition Examination Survey (NHANES) selected representative samples from the U.S. population to assess their health and nutritional status. To ensure ethical standards, the survey was approved by the National Center for Health Statistics. All selected individuals voluntarily provided written informed consent before being included in the study. The relevant data for this study were publicly available through the website (<https://www.cdc.gov/nchs/nhanes/>

[index.htm](#)). Data from multiple NHANES cycles were selected for statistical analysis. A total of 13417 participants were included in our study (Fig. 1).

2.1.1. Variables

Femoral neck bone mineral density (FN BMD) is widely used in clinical practice for OP prediction (Kanis et al., 2008; Wright et al., 2014). Dual-energy X-ray absorptiometry (DXA) is easy to use, with low radiation exposure, and is a common method for BMD measurement (Baran et al., 1997). Blood cadmium levels were directly measured from whole blood samples using mass spectrometry (Caudill et al., 2008). Based on the characteristics of the studied population, the following covariates were selected for analysis. These covariates include sex, age, race, education, marital status, family income-to-poverty ratio, smoking, drinking, hypertension, diabetes, and hyperlipidemia.

2.1.2. Regression analysis

To further assess the relationship between blood cadmium and FN BMD, weighted multiple linear regression analysis was used to examine the association. In 2013–2014, blood analysis indicators were measured for all participants aged 1–11 years, while only half of the participants aged 12 years and older were measured. Therefore, for participants aged 12 and above, "WTSH2YR" was used as the weight variable; participants aged 1–11 in this cycle and other cycles, "WTMEC2YR" was used as the weight variable. To control for confounding factors, we adjusted for different variables and constructed various linear regression models.

2.2. Construction of networks

To further investigate the mechanism by which cadmium affects bone mineral density, we conducted a network toxicology analysis (Chang et al., 2024; He et al., 2024). Initially, we retrieved gene targets related to cadmium exposure from the Comparative Toxicogenomics Database (CTD, <https://ctdbase.org>), a comprehensive platform that

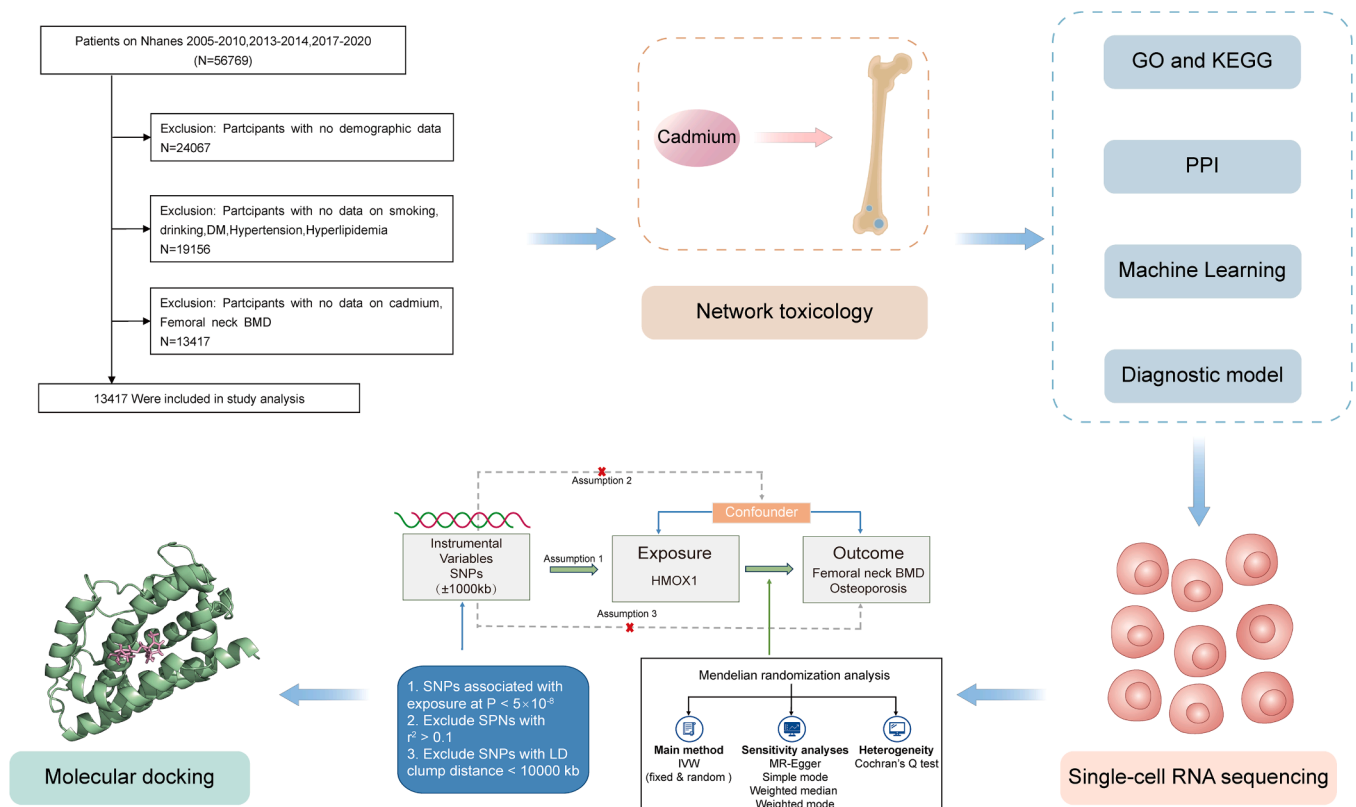


Fig. 1. Overview of the study design.

aggregates biological effect data for chemicals, metals, and other substances (Davis et al., 2023). In this study, we used the keyword “Cadmium” and applied a filtering criterion to retain only genes with more than 10 interactions. OP data were sourced from the Gene Expression Omnibus (GEO) database, specifically bulk datasets GSE7158 and GSE56815 (Lei et al., 2009; Zhou et al., 2018). GEOquery was employed to download and extract both the expression matrix and sample meta-data (Davis and Meltzer, 2007). Expression data were normalized using the normalizeBetweenArrays function from the limma package, followed by log2 transformation. Probe annotations were mapped to gene symbols using platform annotation files (GPL96 and GPL570). Differential expression analysis was performed with the limma package (Ritchie et al., 2015). By intersecting the genes identified in both the CTD and GEO datasets, we ultimately selected key genes for further analysis. Subsequently, we utilized Cytoscape software to construct a “Cadmium - gene - OP” network diagram, visualizing the relationships between cadmium exposure, genes, and OP (Shannon et al., 2003). Enrichment analysis of the intersected genes was carried out using the clusterProfiler package to elucidate their biological relevance in OP (Xu et al., 2024). Gene Ontology (GO) analysis encompassed biological processes (BP), cellular components (CC), and molecular functions (MF). Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis highlighted significant pathways associated with the intersected genes, enhancing our understanding of their potential mechanisms in OP. The Protein-Protein Interaction (PPI) network for the intersected genes was constructed using the STRING database (<https://cn.string-db.org>), employing default parameters and visualizing the network based on degree (Szkarczyk et al., 2023).

2.3. Machine learning

We conducted differential expression analysis by merging GSE7158 and GSE56815 using the limma package in R (Ma and Li, 2024). To assess the predictive power of these genes, we employed machine learning techniques. Specifically, we applied random forest and Support Vector Machine - Recursive Feature Elimination (SVM-RFE) methods to reduce the feature space and identify key genes for classification (Breiman, 2001; Cortes and Vapnik, 1995; Guyon et al., 2002; Liaw and Wiener, 2007). Feature selection was performed using the caret package (Kuhn, 2008), and a random forest model was trained with the randomForest package (Breiman, 2001; Liaw and Wiener, 2007). The importance of variables was determined using the MeanDecreaseGini criterion, with the number of trees set to 1000 to ensure model stability and optimize the tree count to minimize the error rate. Feature selection was further refined through SVM-RFE, which iteratively eliminated less predictive features, retaining only those most informative for the outcome. A logistic regression model was subsequently fitted using the selected genes to evaluate the relationship between gene expression and the disease. The model was trained with the glm function, and odds ratios (OR) with 95 % confidence intervals (CI) were computed for each gene to assess their effects. To validate the robustness of the model, bootstrap resampling with 1000 iterations was performed to estimate model accuracy and calculate the Area Under the Curve (AUC) for performance evaluation. Sensitivity and specificity analyses were also conducted during the bootstrap to assess classification performance across varying threshold settings.

A nomogram was constructed using logistic regression to predict the probability of the outcome, and a calibration curve was plotted to compare predicted and observed probabilities. Additionally, decision curve analysis (DCA) was applied to evaluate the clinical utility of the models (Vickers and Elkin, 2006).

2.4. Single-cell RNA sequencing data analysis

2.4.1. Data acquisition and preprocessing

Single-cell sequencing analysis was conducted using the Seurat

package (version 4.3.0). The raw data from GSE147287 were imported using the Read10X function and stored as a Seurat object using the CreateSeuratObject function (Wang et al., 2021; Zheng et al., 2017). For quality control, cells with low or high feature counts were filtered using subset(nFeature_RNA > 200 & nFeature_RNA < 10,000). Cells with more than 15 % mitochondrial gene content were also removed. The data were then normalized using NormalizeData(normalization.method = “LogNormalize”), and highly variable genes were selected for subsequent analysis using FindVariableFeatures(selection.method = “vst”). Principal component analysis (PCA) was performed with the RunPCA function (Jolliffe, 2002). Clustering was carried out using FindNeighbors(dims = 1:30) and FindClusters(resolution = seq(0.1, 1, 0.2)) to explore the optimal resolution across different parameter settings. UMAP dimensionality reduction was applied and visualized using RunUMAP(dims = 1:15) (McInnes et al., 2018). Cell type annotation was based on known markers from the literature (Wang et al., 2024). The expression of key marker genes was visualized using the DotPlot function.

2.4.2. Cell-cell communication analysis

Cell-cell communication analysis was performed using the CellChat package, utilizing the CellChatDB.human within CellChat, with a focus on secreted signal transduction (Jin et al., 2021; Su et al., 2024). Overexpressed genes and ligand-receptor pairs were identified using the identifyOverExpressedGenes and identifyOverExpressedInteractions functions, respectively. The identified ligand and receptor genes were then mapped to the human protein-protein interaction network to provide a more comprehensive biological interpretation. Communication probabilities between different cell populations were calculated using the computeCommunProb function. Cell communication pathways were inferred with the computeCommunProbPathway function, with rare interactions filtered out. The results were visualized using the netVisual_circle function, which displayed both the number and intensity of interactions. Specific signaling pathways were further analyzed using the netVisual_aggregate function. Lastly, we calculated and visualized the centrality of different cell types within the communication network, showcasing their roles in signal transduction through heatmaps and scatter plots.

2.4.3. GSEA

GSEA was a widely utilized method for identifying biological pathways that were significantly enriched in gene expression data (Subramanian et al., 2005). Differentially expressed genes across various cell subgroups were identified using FindMarkers function. Gene sets for GSEA were obtained from the Molecular Signatures Database (MSigDB, <http://software.broadinstitute.org/gsea/msigdb>) and the literature to explore biological differences between distinct cell subgroups (Liberzon et al., 2015).

2.5. Drug-target MR analysis

2.5.1. Data sources

To further investigate the potential relationship between key genes and OP. We performed Drug-target MR. We used publicly accessible GWAS datasets. The related eQTL data for genes were obtained from eQTLGen (<https://eqtlgen.org/>), which provides data from 31684 European individuals' blood samples (Vosa et al., 2021). The Genetic Factors for Osteoporosis Consortium (GEFOS) is a large international collaborative organization that provides numerous GWAS data related to bone health (Zheng et al., 2015). UK Biobank (UKB) is one of the largest human genetic cohort biobank in the world, and its research data are significant for understanding human health and diseases (Donertas et al., 2021). We selected FN BMD from GEFOS and OP from UKB as outcomes (Table S1). This study followed the STROBE guidelines.

2.5.2. Instrumental variables (IVs) selection

We selected single nucleotide polymorphisms (SNPs) strongly associated with genes to construct IVs ($p < 5.00E-08$), and extracted exposure data based on the gene's location, choosing SNPs within the ± 1000 kb range of each gene's transcription start site (TSS). We removed linkage disequilibrium based on the European population reference genome, setting r^2 to 0.1 and kb to 10,000. The F -statistic can be used to assess the strength of the IVs (Bowden et al., 2019). We calculated the value using $F = \beta^2 / \text{se}^2$. The selection process follows the three main assumptions of MR: (1) a strong and robust association between IVs and exposure; (2) IVs are independent of confounding factors affecting the exposure-outcome relationship; (3) genetic variation affects the outcome only through the exposure and not through other pathways (Emdin et al., 2017).

2.5.3. MR and Meta-analysis

MR analysis was conducted using the TwoSampleMR package. This study used five MR methods to explore the relationship between genes and diseases. Inverse Variance Weighted (IVW) method assumes that all instrumental variables are valid and that there is no horizontal pleiotropy, providing high statistical power (Slob and Burgess, 2020). Therefore, we selected the IVW as the primary analytical method. Heterogeneity was assessed using Cochran's Q test (Greco et al., 2015). Pleiotropy refers to the potential of a gene or genetic variation to have more than one independent phenotype. Horizontal pleiotropy was tested using MR-Egger regression intercept (Bowden et al., 2015). Leave-one-out sensitivity analysis was applied to assess the robustness of the results (Burgess et al., 2017). To obtain more stable results, we used the meta package to perform a meta analysis of the IVW estimates from different cohort MR analyses and visualized the results using the forest function (Schwarzer et al., 2015).

2.6. Molecular docking and Molecular dynamic simulation

2.6.1. Candidate drug prediction

The Drug Signatures Database (<http://dsigdb.tanlab.org/>) collected a large number of drug-related genes and compounds (Yoo et al., 2015). We utilized the database to evaluate whether the target genes could serve as drug targets and predict the potential effects of the drugs.

2.6.2. Molecular docking

To evaluate the binding energy and interaction patterns between the candidate drug and its target. Molecular docking was performed using AutoDock Vina 1.1.2 (Trott and Olson, 2010). We obtained the protein crystal structure from the PubChem Compound Database (<https://pubchem.ncbi.nlm.nih.gov/>). The structural data of the drug were obtained from the Protein Data Bank (PDB, <http://www.rcsb.org/>). We prepared the protein using PyMol 1.8.5 software, including hydrogenation, removal of water molecules, etc., and converted the PDB format of small molecules and receptor proteins into PDBQT format (Delano, 2002). During docking, the conformation with the lowest binding energy was selected as the docking conformation, and visualization analysis was performed using Plip 2021 (Adasme et al., 2021).

2.6.3. Molecular dynamics simulations

To further assess the stability and interactions of the molecular docking, we performed molecular dynamics simulations using AMBER 22 (Salomon-Ferrer et al., 2013). The charges of small molecule were calculated using the antechamber module and the Hartree-Fock (HF) SCF/6-31 G* method in Gaussian 09 software. We selected the TIP3P water model and added sodium and chloride ions to the complex to neutralize the charges. The system was optimized using the steepest descent method and the conjugate gradient method. Following energy optimization, the system was heated for 200 ps. The system was maintained at a temperature of 298.15 K and underwent a 500 ps canonical ensemble (NVT). A 500 ps equilibration simulation under

isothermal-isobaric ensemble (NPT) conditions was performed for the entire system. Finally, both complex systems underwent 100 ns NPT under periodic boundary conditions. The system pressure was set to 1 atm, the integration time step was 2fs, and the trajectory was saved every 10 ps for subsequent analysis.

2.7. Statistical analysis

All statistical analyses were performed using R version 4.1.3. For continuous variables, group differences were assessed using Student's t -test or the Wilcoxon rank-sum test. For categorical variables, group associations were assessed using chi-square tests or Fisher's exact test based on the expected frequency of each category. A p -value less than 0.05 was considered statistically significant. The data used in this study are publicly available and have received ethical approval.

3. Results

3.1. Cross-Sectional analysis

FN BMD was grouped by quartiles (Table 1). Notably, participants with lower FN BMD tended to have higher blood cadmium levels. We further employed multiple linear regression models and found a significant negative correlation between blood cadmium and FN BMD, with this significance persisting after adjusting for confounders. Specifically, with an increase in blood cadmium, FN BMD decreased by 0.006 g/cm² (-0.011 , -0.001 , $p = 0.017$). These results suggested a significant negative correlation between blood cadmium levels and BMD, indicating that cadmium exposure was an important factor in reducing BMD (Table 2).

3.2. Identification of blood cadmium targets affecting OP

We selected cadmium-related genes with "Interactions > 10" from the CTD. OP-related genes were obtained from the GEO. A total of 24 significantly different genes were identified (Fig. 2A). To further explore the potential biological functions of key genes, we performed GO and KEGG enrichment analysis. We found that these genes are involved in the following BP: response to cadmium ion, response to oxidative stress, and cellular response to chemical stress. CC included transcription repressor complex, nuclear envelope, mitochondrial outer membrane, and others. MF analysis revealed RNA polymerase II-specific DNA-binding transcription factor binding, DNA-binding transcription factor binding, cholesterol transfer activity, and others. KEGG analysis showed that the genes were enriched in pathways such as endocrine resistance, mineral absorption, and lipid and atherosclerosis (Fig. 2B). Based on the results of GO and KEGG enrichment, we found that these genes are closely related to immune response, oxidative stress, and other processes, providing clues for further investigating their potential roles in diseases. Proteins such as AKT1, JUN, TP53, EGFR, and HMOX1 showed tight interactions with other proteins in the network (Fig. 2C). In OP, FOXO3, MAP1LC3B, and MT1G were significantly upregulated, while CCND1 and HMOX1 were downregulated (Fig. 2D-F). Correlation analysis indicated relationships between multiple genes (Fig. 2G). We identified key genes associated with cadmium exposure and revealed their potential roles such as immune response and oxidative stress, providing important clues for further investigation of their functions in related diseases.

3.3. Machine learning identifies key genes

To identify key genes from another perspective, we employed machine learning approaches. Random forest analysis showed that the genes with the highest MeanDecreaseGini scores included CCND1, FOXO3, SIRT1, MT1G, HMOX1, MAP1LC3B, HSP90B1, TP53, EGFR, and XAF1 (Fig. 3A, B). The intersection of top genes from random forest

Table 1
Characteristics of the participants.

Characteristics		Q1	Q2	Q3	Q4	p Value
Age (years)		61.68 ± 0.36	53.54 ± 0.40	48.24 ± 0.35	43.09 ± 0.38	< 0.0001
Blood cadmium (ug/L)		0.58 ± 0.02	0.55 ± 0.01	0.55 ± 0.02	0.51 ± 0.01	0.01
Sex (%)						< 0.0001
	Male	1171(30.10)	1637(46.78)	1849(54.73)	2134(63.71)	
	Female	2212(69.90)	1719(53.22)	1495(45.27)	1200(36.29)	
Race (%)						< 0.0001
	Black	349(4.40)	487(6.42)	688(10.48)	1071(18.37)	
	White	2054(80.10)	1778(76.20)	1550(70.89)	1244(61.61)	
	Mexican American	399(4.24)	543(6.30)	599(8.15)	563(9.06)	
	Other	581(11.26)	548(11.08)	507(10.48)	456(10.97)	
Marital status (%)						< 0.0001
	Married/Living with Partner	1887(60.32)	2146(68.56)	2175(69.67)	2070(65.88)	
	Never married	255(7.39)	330(8.66)	491(12.92)	713(20.51)	
	Widowed/Divorced/Separated	1241(32.29)	880(22.78)	678(17.41)	551(13.61)	
Family income-to-poverty ratio (%)						< 0.001
	Low income	936(17.14)	921(15.79)	924(17.52)	968(19.36)	
	Middle income	1387(38.27)	1221(33.39)	1260(32.79)	1239(35.20)	
	High income	1060(44.60)	1214(50.83)	1160(49.69)	1127(45.44)	
Education (%)						0.02
	Less Than 9th Grade	412(5.74)	337(4.41)	346(5.23)	302(5.22)	
	9–11th Grade	470(10.73)	470(10.15)	456(9.51)	555(12.85)	
	High School Grad/GED or Equivalent	835(24.82)	771(23.49)	805(25.50)	800(24.82)	
	Some College or AA degree	913(28.79)	976(30.98)	970(30.07)	968(30.02)	
	College Graduate or above	753(29.92)	802(30.97)	767(29.69)	709(27.09)	
Drinking (%)						< 0.0001
	Former	648(15.94)	574(13.60)	520(12.86)	447(11.46)	
	Never	626(14.15)	404(9.74)	351(8.20)	295(7.27)	
	Mild	1276(43.04)	1244(39.49)	1156(37.89)	1116(35.61)	
	Moderate	467(15.70)	543(19.15)	527(17.96)	542(17.13)	
	Heavy	366(11.18)	591(18.02)	790(23.09)	934(28.53)	
Smoking (%)						< 0.0001
	Former	1017(29.49)	1002(29.60)	888(27.17)	748(23.53)	
	Never	1722(52.78)	1595(48.70)	1606(48.72)	1707(50.92)	
	Now	644(17.73)	759(21.69)	850(24.12)	879(25.55)	
Hypertension (%)						< 0.0001
	No	1503(51.59)	1789(57.09)	1962(60.85)	2055(64.82)	
	Yes	1880(48.41)	1567(42.91)	1382(39.15)	1279(35.18)	
Diabetes Mellitus (%)						0.36
	No	2359(76.45)	2417(76.79)	2492(78.15)	2450(76.52)	
	IFG	169(4.62)	154(4.56)	165(4.89)	166(5.33)	
	IGT	183(4.38)	128(3.19)	118(3.62)	128(3.97)	
	Yes	672(14.54)	657(15.47)	569(13.34)	590(14.18)	
Hyperlipidemia (%)						< 0.0001
	No	701(21.80)	846(24.90)	899(26.65)	968(29.72)	
	Yes	2682(78.20)	2510(75.10)	2445(73.35)	2366(70.28)	

Note :
Mean ± SD for continuous variables; the p-value was calculated by the weighted linear regression model. (%) for categorical variables: the p-value was calculated by the weighted chisquare test.
GED, general educational development; IFG, Impaired Fasting Glycaemia; IGT, Impaired Glucose Tolerance.

Table 2
Associations between blood cadmium and BMD.

	Model 1		Model 2		Model 3	
Variables	β (95 %CI)	p Value	β (95 %CI)	p Value	β (95 %CI)	p Value
Femoral neck BMD	−0.011 (−0.017, −0.005)	< 0.001	−0.012(−0.017, −0.008)	< 0.0001	−0.006(−0.011, −0.001)	0.017

Note :
Model 1-Unadjusted.
Model 2-Model 1 additionally adjusted for the age, sex, race, marital status, family income-to-poverty ratio, and education.
Model 3-Model 2 plus additional adjustment for drinking, smoking, hypertension, diabetes mellitus, and hyperlipidemia.
BMD, bone mineral density.

and SVM-RFE algorithms highlighted six common genes: FOXO3, CCND1, MAP1LC3B, HMOX1, EGFR, and MT1G (Fig. 3C). Multivariable logistic regression further confirmed that FOXO3, MAP1LC3B, and MT1G were positively correlated with OP, while CCND1 and HMOX1 were negatively correlated with OP (Fig. 3D). Internal validation conducted through bootstrap analysis showed that the model had stable sensitivity and specificity (Fig. 3E–H). A nomogram was constructed to visualize the model, and calibration curves indicated robust predictive

performance (Fig. 3I, J). DCA showed that the combination of these genes outperformed the use of any single gene in enhancing clinical decision-making (Fig. 3K).

3.4. Single-cell RNA sequencing of HMOX1 in macrophages

Single-cell RNA sequencing of OP identified macrophages, monocytes, b cells, and other subpopulations (Fig. 4A, B). Among the key

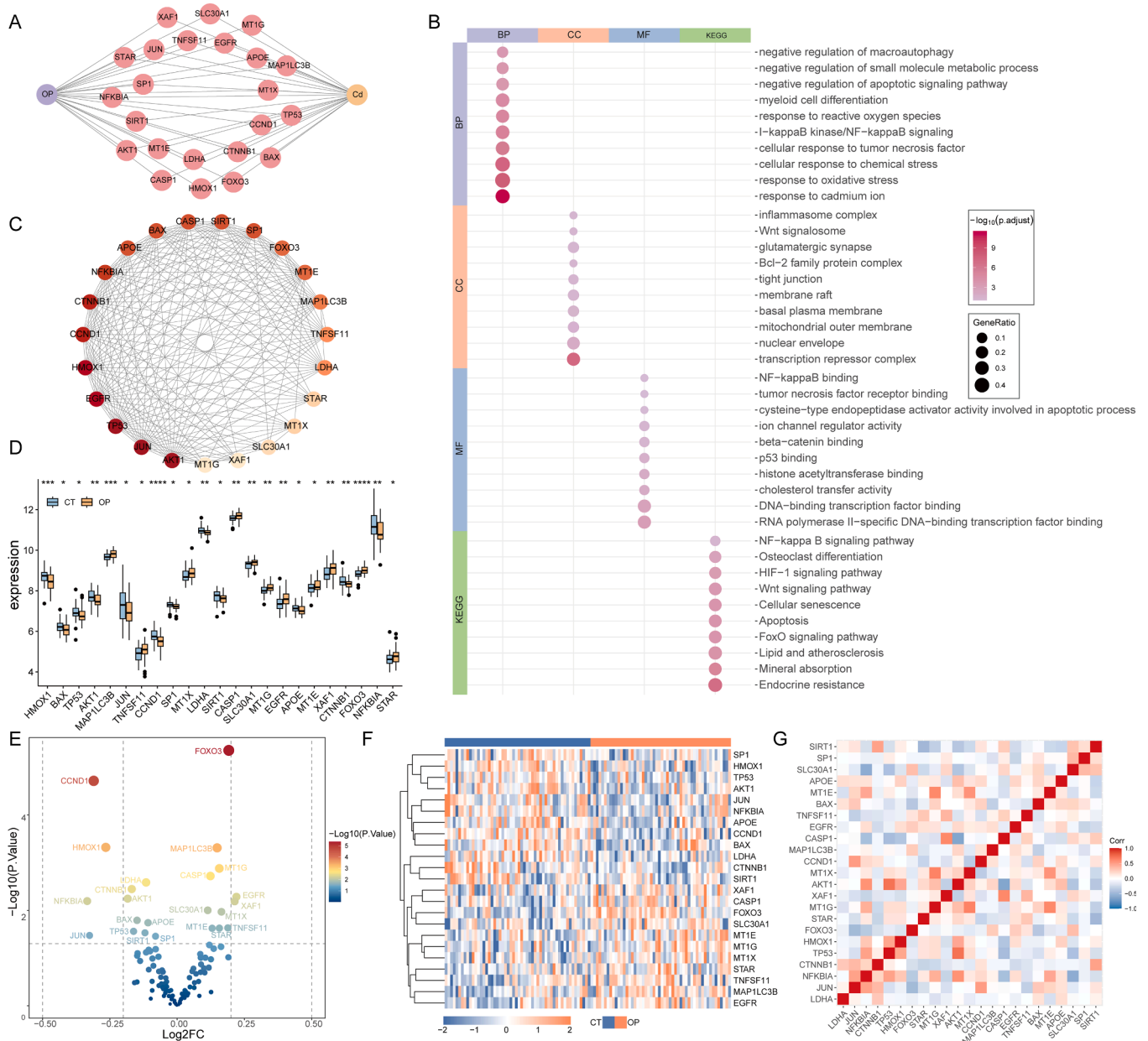


Fig. 2. Network Toxicology for Cadmium and OP. (A) Target genes between cadmium and OP. (B) Enrichment results of GO and KEGG. The circle size reflects the proportion of cells expressing each gene, and the color intensity indicates the level of gene expression. (C) The PPI network for target genes. Greater importance is represented by a darker color. (D) Boxplots showing the expression levels of target genes in the OP and control (CT) groups. Significance is indicated as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ and **** $p < 0.0001$. (E) Volcano plot of target genes. Upregulation is represented in red, while downregulation is shown in blue. (F) Heatmap of target genes. Upregulation is represented in orange, while downregulation is shown in blue. (G) Correlation analysis of the expression levels of target genes. Strong positive correlations are shown in darker red, and strong negative correlations are shown in darker blue.

genes selected by machine learning, we found that HMOX1 was most significantly expressed in macrophages (Fig. 4C, D). Therefore, we stratified Macrophages into HMOX1-positive (HMOX1 + Macr) and HMOX1-negative (HMOX1- Macr) subpopulations. HMOX1 + Macr had stronger interactions with DC, B cells, NK/T cells, and monocytes (Fig. 4E, S1), indicating that HMOX1 mediated intercellular communication. Notably, ANXA1 and MIF signaling pathways were enriched in HMOX1 + Macr, potentially driving macrophage polarization. Pathways such as LGALS9 and TNFSF13B suggested their involvement in immune regulation (Fig. 4F). Pathway analysis highlighted increased activity of pathways such as SPP1, ANNEXIN, and GRN in HMOX1 + Macr (Fig. 4G). HMOX1 + Macr had higher incoming and outgoing interaction strength (Fig. 4H). This indicated that HMOX1 may

regulate the function of macrophages and affect their interactions with other cells. Notably, similar to cell communication, GSEA also found that HMOX1 + Macr was associated with the upregulation of M2 macrophage polarization (Fig. 4I). These results suggested that HMOX1 was closely related to the function and intercellular interactions of macrophages and affected macrophage polarization.

3.5. Drug-target MR study between HMOX1 and OP

3.5.1. Preliminary analysis

To further explore the potential role of HMOX1 in OP, we performed Drug-target MR. After excluding SNPs using strict criteria, we selected IVs that met the standards. The F -values of all IVs were greater than 10,

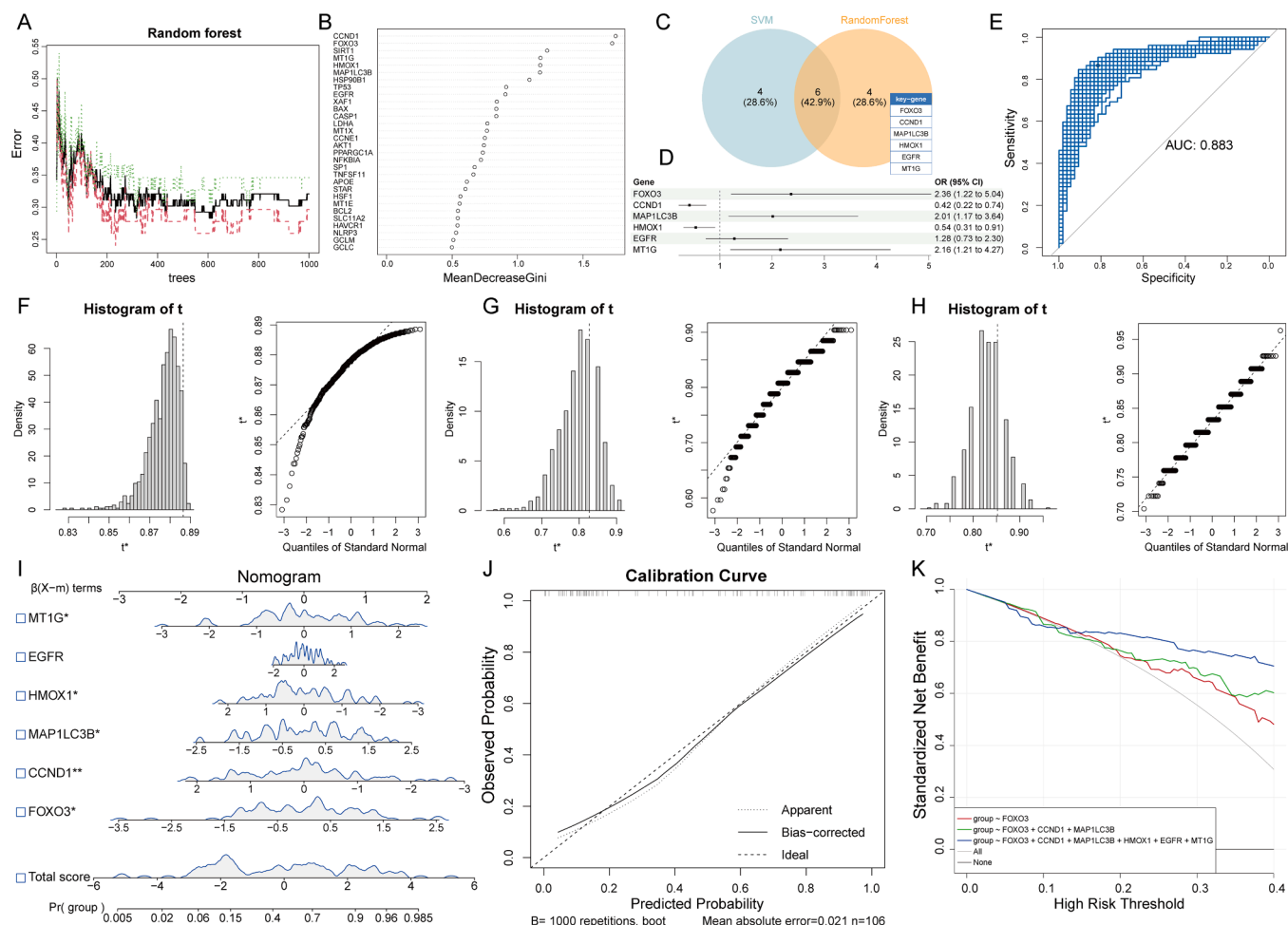


Fig. 3. Machine Learning Identifies Key Genes. (A) The curve illustrates the error rate of the Random Forest model as the number of decision trees increases. The x-axis represents the number of trees, while the y-axis denotes the corresponding error. (B) Ranking of gene importance in the Random Forest model based on the decrease in the Gini index. (C) Venn diagram indicating that the key genes FOXO3, CCND1, MAP1LC3B, HMOX1, EGFR and MT1G are identified in both the SVM model and the Random Forest model. (D) Odds ratios (OR) and 95 % confidence intervals for the key genes, demonstrating the strength of their association with the model. (E) Internal validation of the model using the receiver operating characteristic (ROC) curve, with an area under the curve (AUC) of 0.883. (F) Distribution of the AUC values for the model. (G) Sensitivity analysis of the AUC curve. (H) Specificity analysis of the AUC curve. (I) Nomogram displaying the contribution of FOXO3, CCND1, MAP1LC3B, HMOX1, EGFR and MT1G to the total risk score, used for outcome prediction. (J) Calibration curve comparing the predicted probabilities with the observed outcomes, with the mean absolute error reflecting the calibration accuracy of the model. (K) Decision curve analysis (DCA) of the key genes (FOXO3, CCND1, MAP1LC3B, HMOX1, EGFR and MT1G), demonstrating that the combination of these genes outperforms individual genes in improving clinical decision-making.

indicating that there were no weak instrumental variables (Table S2-S5). We found that HMOX1 increased FN BMD ($\beta=0.0399$, 95 % CI: 0.0076–0.0722, $p=1.54\text{E-}02$) and acted as a protective factor for OP (OR=0.9989, 95 % CI: 0.9979–0.9998, $p=2.32\text{E-}02$), which was consistent with the results of multivariable logistic regression.

3.5.2. Replication and meta-analysis

We performed the analysis in the replication cohort in the same manner to increase the stability of the results (Table S6, S7). We also found that HMOX1 is a protective factor for OP (OR=0.9986, 95 % CI: 0.9974–0.9997, $p=1.46\text{E-}02$). The study results showed no heterogeneity or pleiotropy (Table S8, S9). Leave-one-out analysis, scatter plots, and funnel plots supported the robustness of the MR estimates (Fig. 5A–I). Meta-analysis further revealed that HMOX1 is a protective factor for OP (Fig. 5J). These results further suggest from a genetic perspective that HMOX1 may have a potential protective role in OP.

3.6. Candidate drug prediction

3.6.1. Molecular docking

Based on the results of the drug enrichment analysis, we found that geniposide can target HMOX1. Geniposide and HMOX1 were selected for docking, with a binding energy of -6.9 (kcal/mol), suggesting a good docking result. Hydrogen bonds indicate the binding strength between the ligand and the protein. Geniposide and HMOX1 formed hydrogen bonds with HIS-25, TYR-134, SER-142, LYS-179, and ARG-183. These interactions formed the basis for the binding of the small molecule and the protein (Fig. 6A).

3.6.2. Molecular dynamics simulations

To further improve the accuracy of the simulation, we performed molecular dynamics simulations based on molecular docking to capture the dynamic behavior and stability of the complex more precisely. We found that the complex reached a stable state after 20 ns, with the root mean square deviation (RMSD) fluctuation range smaller than $1 \text{ \AA}$, indicating good stability of the complex (Fig. 6B). The root mean square fluctuation (RMSF) showed an overall fluctuation range between $0.5 \text{ \AA}$

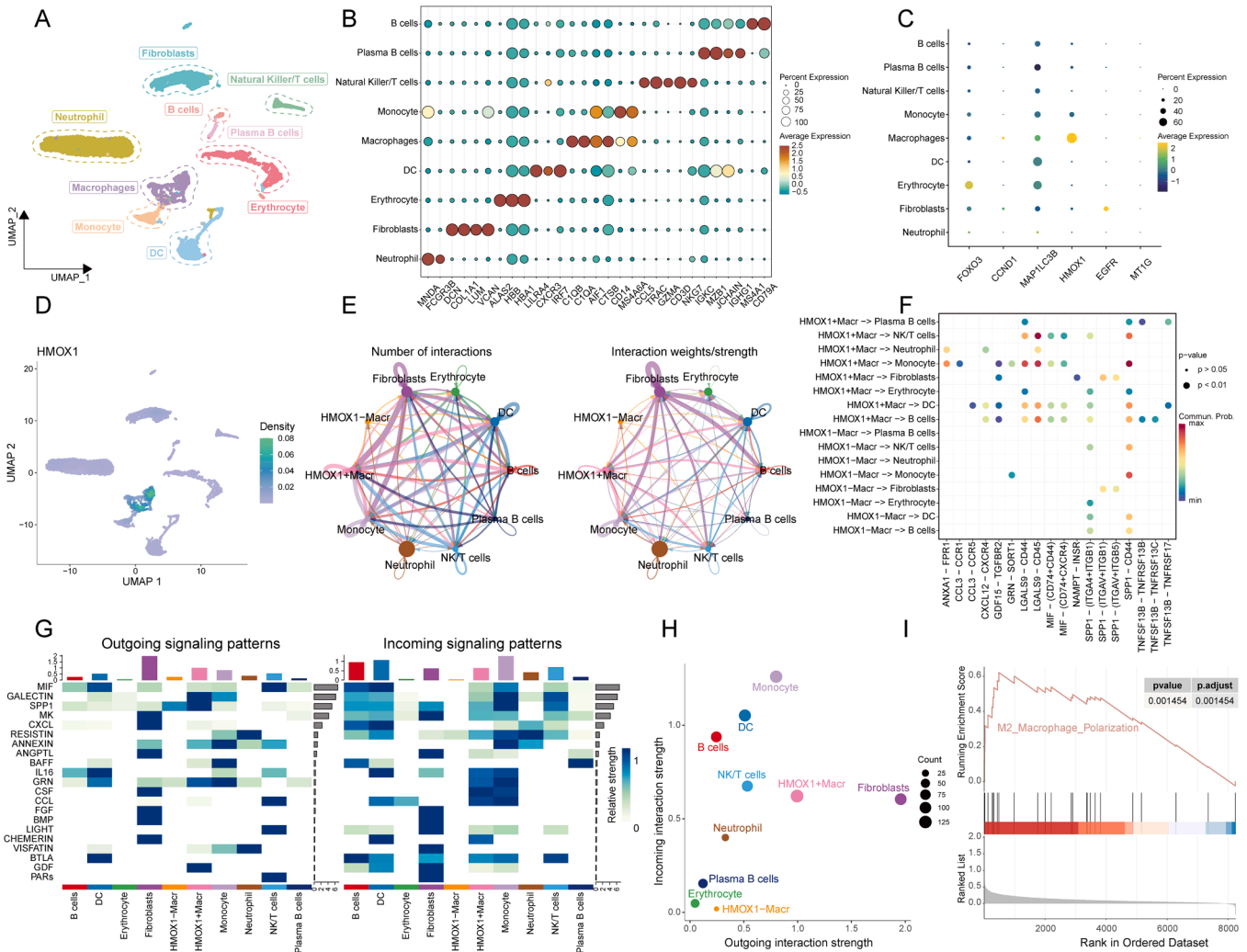


Fig. 4. Single-cell RNA sequencing Analysis of Key Genes in OP. (A) UMAP plot of different cell clusters, indicated by various colors. (B) Single-cell annotation information. (C) Expression differences of key genes (FOXO3, CCND1, MAP1LC3B, HMOX1, EGFR and MT1G) in different cell types. The size of the circles represents the proportion of cells expressing the gene, while the color intensity indicates expression levels. (D) Expression of HMOX1 in UMAP plot. The color intensity indicates the density. (E) Left panel: the quantity of interactions between various cell types, with thicker lines represent a higher number of interactions. Right panel: the intensity of communication between cell types, with thicker lines signifying stronger interactions. (F) Communication probability and signaling pathways between HMOX1 + and HMOX1- Macr and other cell types. Red indicates higher probability, blue indicates lower probability, and larger circles indicate smaller *P*-value. (G) Signaling patterns divided into outgoing and incoming categories. Darker colors indicate higher relative signaling strength, while lighter colors indicate lower relative strength. (H) Comparison of interaction strength. The x-axis represents the outgoing interaction strength, and the y-axis represents the incoming interaction strength. The color indicates cell types and the size of the circles represents the strength. (I) GSEA of differential genes between the HMOX1 + and HMOX1- Macr. Score > 0 indicates that HMOX1 + Macr are associated with upregulation of pathway.

and 2 Å, with two distinct peaks between residues 0–50, indicating higher activity, but no significant conformational changes occurred (Fig. 6C). The radius of gyration (RoG) fluctuation was also stable, further indicating the compact structure of the complex (Fig. 6D). The binding energy calculated using the molecular mechanics generalized born surface area (MM-GBSA) method was -22.78 (kcal/mol) (Table 3). The top ten amino acids contributing to the binding of the protein-ligand complex are shown in Fig. 6E. The docking complex exhibited a stable hydrogen bond pattern (Fig. 6F). Additionally, we extracted the complex conformations at every 10 ns interval during the 0–100 ns process (Figure S2). In conclusion, our results showed that the complex exhibited minimal conformational changes during the simulation and remained stable, indicating that geniposide has the potential to target and bind to HMOX1.

4. Discussion

This study aimed to elucidate the key role and biological mechanisms of cadmium exposure in the pathogenesis of OP. Through a cross-sectional study of NHANES, we found a significant negative correlation between blood cadmium and BMD. Network toxicology was performed by integrating the CTD and GEO, and a PPI network was constructed. Key genes were further identified through machine learning, and their expression in specific cells was observed using single-cell sequencing analysis. We found that the expression of HMOX1 was significant in macrophages. Cell communication analysis revealed that HMOX1 + Macr had higher communication activity. It was found that HMOX1 + Macr were associated with the upregulation of M2 macrophage polarization through GSEA. MR demonstrated the potential protective effect of HMOX1 in OP at the genetic level. Furthermore, molecular docking and molecular dynamics simulations showed that geniposide could specifically target and bind to HMOX1, offering

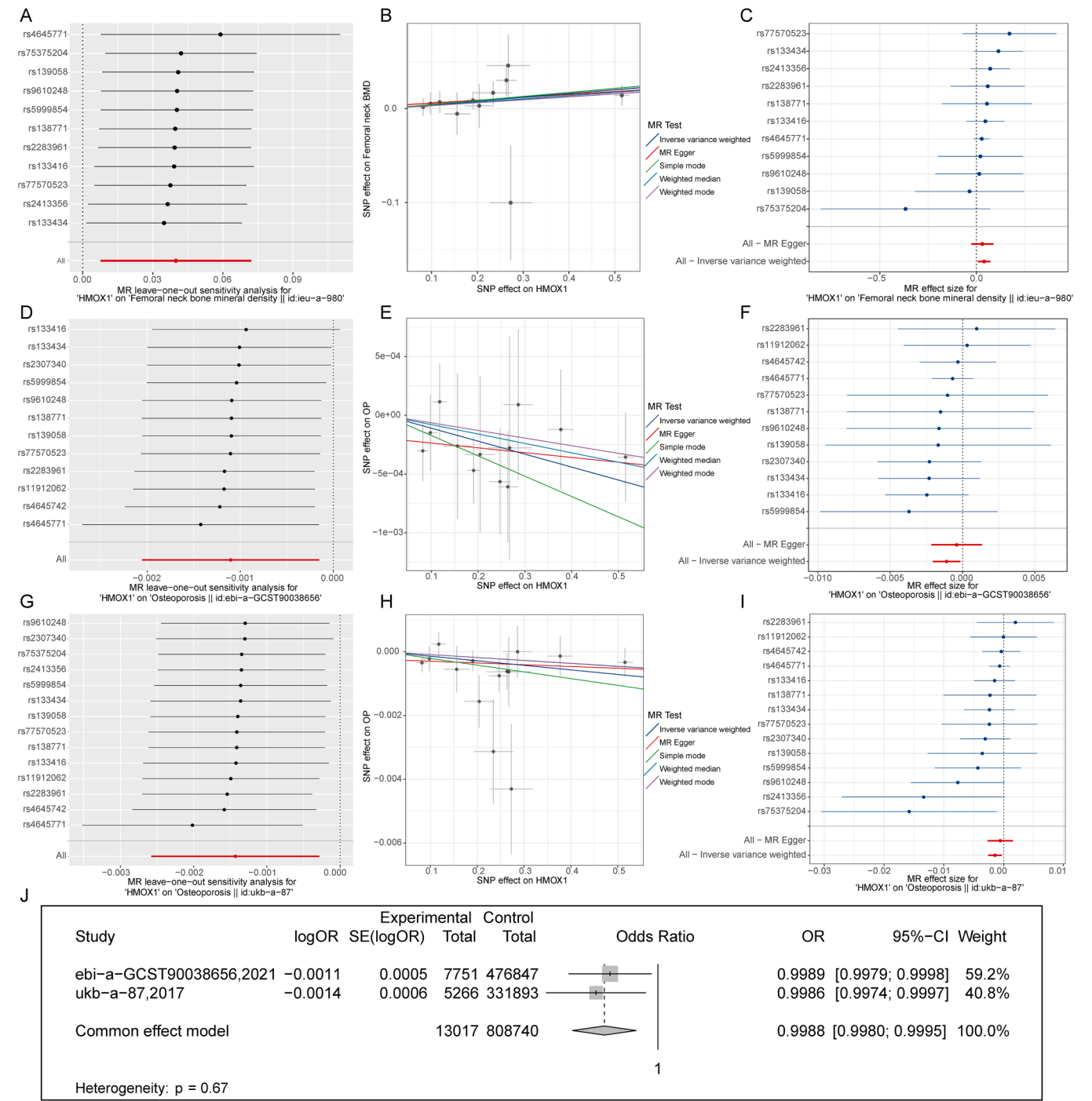


Fig. 5. Durg-target MR and Meta analysis. (A)Leave-one-out plot of HMOX1 on FNBMD. (B)Scatter plot of HMOX1 on FNBMD. (C)Forest plot of HMOX1 on FNBMD. (D)Leave-one-out plot of HMOX1 on OP in the preliminary analysis. (E)Scatter plot of HMOX1 on OP in the preliminary analysis. (F)Forest plot of HMOX1 on OP in the preliminary analysis. (G)Leave-one-out plot of HMOX1 on OP in the replication analysis. (H)Scatter plot of HMOX1 on OP in the replication analysis. (I)Forest plot of HMOX1 on OP in the replication analysis. (J)Meta analysis of different OP cohorts.

potential directions for intervention therapy.

Cadmium is a commonly found toxic heavy metal in the environment, and it has been shown to be closely related to decreased BMD and the onset of OP. Based on the representative sample in NHANES, we found a significant negative correlation between blood cadmium and FN BMD, but the biological mechanisms behind this process remain unclear, with inflammation being a widely discussed factor. An observational study based on electroplating workers found that serum levels of IL-6 and TNF- α were significantly higher in the cadmium exposure group compared to the non-exposure group, suggesting that cadmium

exposure is closely related to inflammation (Ramadan and Saif Eldin, 2022). Cadmium exposure depletes glutathione (GSH) and disrupts the binding of NRF2 with the ARE (Chou and Tsai, 2023). A study has shown that cadmium exposure induces oxidative stress in osteoblasts by inhibiting the NRF2/NQO1 pathway (Jia et al., 2023). In the absence of NRF2, L-NRF1 is compensatorily upregulated, which subsequently increases the expression of NFATc1, a critical transcription factor in osteoclast differentiation (Liu et al., 2024). The overexpression of NFATc1 enhances RANKL expression and activates NF- κ B, thus promoting osteoclastogenesis. HMOX1 has been shown to alleviate this

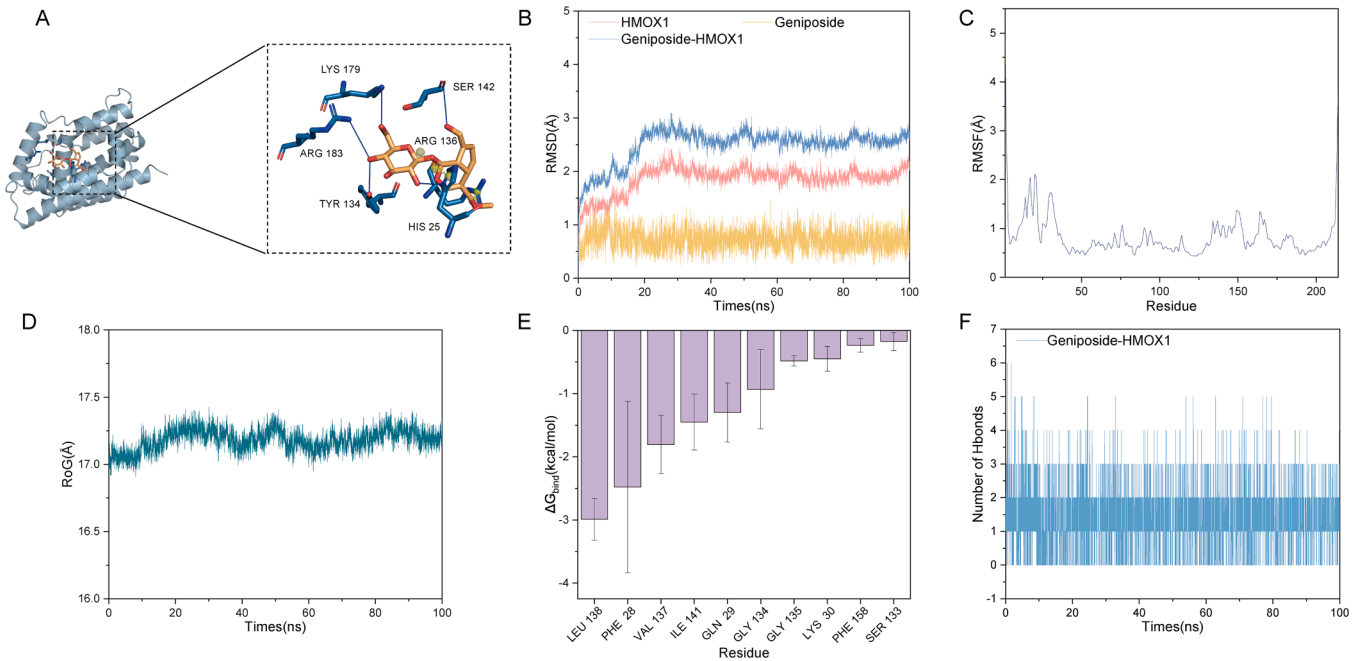


Fig. 6. Molecular Docking and Molecular Dynamic Simulation between Geniposide and HMOX1. (A) Binding mode of geniposide and HMOX1. The left panel shows the overall binding pose, while the right panel illustrates the interaction details with residues. The blue lines represent hydrogen bonds, and the yellow dashed lines indicate salt bridges. B-D) RMSD, RMSF, RoG of the complex. Smaller fluctuations suggest a more stable structure. (E) The top 10 residues contributing to the binding affinity of the complex. (F) Variation in the number of hydrogen bonds of the complex during molecular dynamics simulations.

Table 3

Binding free energies and energy components predicted by MM/GBSA (kcal/mol).

System name	Geniposide/HMOX1
ΔE_{vdw}	-27.23 ± 2.19
ΔE_{elec}	-11.92 ± 4.21
ΔG_{GB}	20.18 ± 4.16
ΔG_{SA}	-3.80 ± 0.25
ΔG_{bind}	-22.78 ± 2.82

ΔE_{vdw} : van der Waals energy.

ΔE_{elec} : electrostatic energy.

ΔG_{GB} : electrostatic contribution to solvation.

ΔG_{SA} : non-polar contribution to solvation.

ΔG_{bind} : binding free energy.

process (Tan et al., 2022). It is important to note that activated NF- κ B also promotes the expression of NFATc1, establishing a positive feedback loop that further accelerates osteoclast differentiation, with reactive oxygen species (ROS) playing a crucial role in this mechanism. Even low levels of cadmium exposure (blood cadmium $<2.40 \mu\text{g/L}$) may disrupt redox balance by reducing antioxidant enzyme activity and increasing lipid peroxidation, leading to an increase in ROS, oxidative stress, activation of NF- κ B and IL-8 release, and induction of IL-6 and TNF- α release (Freitas and Fernandes, 2011; Goyal et al., 2021). To further explore its specific mechanisms in OP, we conducted network toxicology and identified 24 key genes. Enrichment analysis also revealed that these genes were closely related to NF- κ B signaling, inflammatory response, and oxidative phosphorylation in OP. Further single-cell sequencing analysis revealed that HMOX1 was most significant in macrophages.

HMOX1 is a key enzyme involved in heme catabolism and can produce biliverdin and carbon monoxide (Naito et al., 2014). HMOX1 is induced under the stimulation of various stressors, such as oxidative stress and inflammatory cytokines. NRF2 dissociates from KEAP1 in response to stress stimuli, translocates to the nucleus, and interacts with small musculoaponeurotic fibrosarcoma (sMAF) proteins to recognize

and bind the ARE sequence, thereby positively promoting the transcription of HMOX1 (Che et al., 2021). Cells lacking HMOX1 exhibit increased sensitivity to oxidative damage (Tanaka et al., 2011). NRF2 activators have been utilized to enhance HMOX1 activity, thus maintaining redox homeostasis. BACH1 has been shown to compete with NRF2 for binding to the HMOX1 promoter region, inhibiting its expression and negatively regulating the NRF2 signaling pathway. NF- κ B plays a crucial role in bone metabolism; its activation inhibits osteoblast differentiation and promotes osteoclastogenesis, thereby accelerating bone resorption. HMOX1 and its metabolites inhibit the NF- κ B signaling pathway through several mechanisms. CO and biliverdin, produced during heme metabolism, play key roles in this process. CO inhibits RANKL-induced osteoclast differentiation by suppressing the ROS/IKK/NF- κ B signaling pathway (Bak et al., 2017). Additionally, CO interferes with TLR4 transport and restricts the translocation of NF- κ B to the nucleus (Watabe et al., 2025). CO also induces p65 S-glutathionylation, leading to NF- κ B inactivation (Yeh et al., 2014). CO can activate the NRF2 signaling pathway, induce HMOX1 expression, create a positive feedback loop, and promote M2 polarization in macrophages (Jin et al., 2022). Biliverdin is converted to bilirubin by biliverdin reductase; both compounds participate in ROS clearance and inhibit oxidative stress-induced NF- κ B activation (Huang et al., 2022). Both biliverdin and bilirubin suppress IL-18-mediated NF- κ B activation, and bilirubin further inhibits NF- κ B pathway activation by blocking p65 phosphorylation (Li et al., 2020; Zabalgoitia et al., 2008). The schematic diagram of the signaling pathways is shown in Fig. 7. HMOX1 has anti-inflammatory protective effects, and the absence of HMOX1 leads to elevated levels of circulating inflammatory markers such as TNF- α (Chudy et al., 2024). Pre-treatment with CO or bilirubin significantly reduces IL-1B release from macrophages, suggesting the potential role of HMOX1 in inflammatory regulation (Vitali et al., 2020). Activation of the NRF2/HMOX1 pathway can effectively improve inflammation and cell apoptosis induced by heavy metal mixtures (Eddie-Amadi et al., 2022). HMOX1 can inhibit the activity of iNOS (inducible nitric oxide synthase) in macrophages and is associated with an anti-inflammatory macrophage phenotype (Townsend et al., 2014).

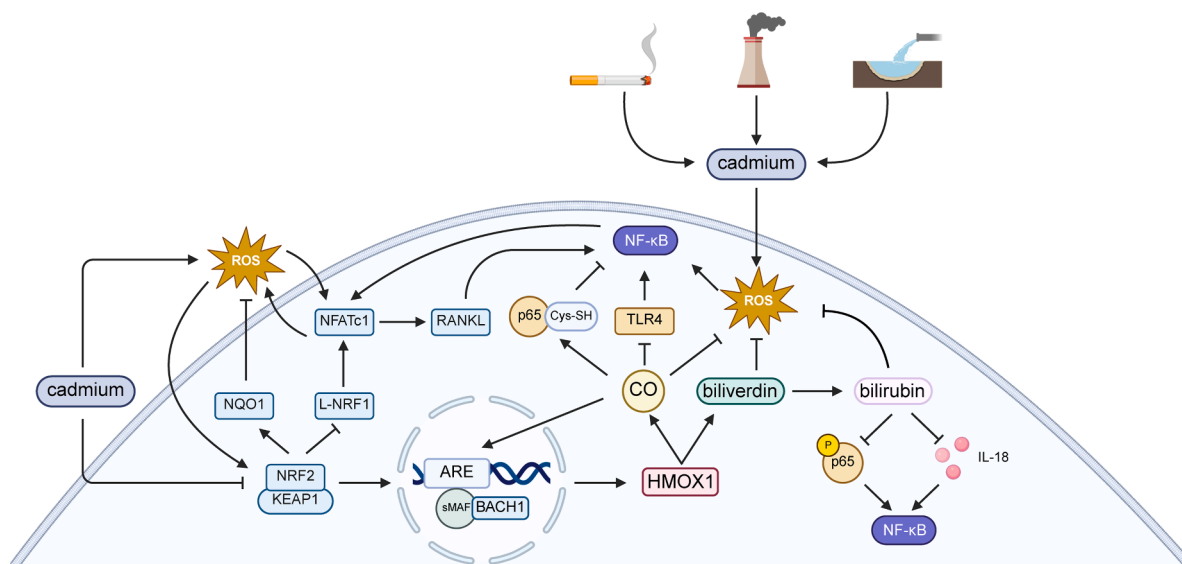


Fig. 7. Summary diagram of HMOX1-mediated regulation of cadmium-induced OP via NF- κ B and NRF2 signaling pathways. Cadmium exposure induces oxidative stress by promoting the production of ROS, which activate the NF- κ B signaling pathway and trigger inflammatory responses. ROS promote the release of NRF2 from KEAP1, facilitating its translocation into the nucleus, where it binds to ARE and initiates the transcription of HMOX1. HMOX1 catalyzes the degradation of heme into CO and biliverdin, both of which collaboratively modulate oxidative stress. CO inhibits NF- κ B activity by regulating TLR4 and the S-glutathionylation of p65. Biliverdin is further reduced to bilirubin, which suppresses NF- κ B activation by downregulating p65 phosphorylation and IL-18. Cadmium exposure inhibits the NRF2/NQO1 pathway, impairs the antioxidant defense response, and facilitates osteoclastogenesis via the NFATc1/RANKL axis, thereby exacerbating OP. Figure created with BioRender (<http://biorender.com>).

Other research further revealed that HMOX1 not only promotes macrophage polarization towards the M2 subtype but also suppresses pro-inflammatory responses, stimulates IL-10 secretion, and boosts the antioxidant capacity of macrophages (Lv et al., 2023). HMOX1 expression was positively correlated with M2 polarization-related markers and could serve as a therapeutic strategy to enhance M2 polarization (Ye et al., 2021). Regulating the M2/M1 macrophage ratio under the NF- κ B pathway and inhibiting oxidative stress through the NRF2/HMOX1 pathway could effectively alleviate periodontitis and promote new bone formation (Yin et al., 2025). This is similar to our findings. We found that HMOX1 + Macr was associated with the upregulation of M2 macrophage polarization, while HMOX1 was downregulated in OP, suggesting that macrophages under cadmium exposure may have stronger pro-inflammatory characteristics, which means that cadmium exposure may promote bone metabolic imbalance and exacerbate OP progression by regulating macrophage polarization.

HMOX1, as a potential target, plays an important regulatory role in the occurrence and progression of various diseases. A multi-omics study suggested that HMOX1 could be a drug target for upper and lower respiratory diseases (Wang et al., 2025). However, its role in OP remains unclear. We found that HMOX1 has a protective effect in OP through drug-target MR. Further molecular docking showed that geniposide can target and bind to HMOX1, and molecular dynamics simulations confirmed the stability of the binding. Geniposide, an iridoid glycoside derived from the fruit of *Gardenia jasminoides* (Xiaofeng et al., 2012), has garnered attention for its potential role in the prevention and treatment of OP. Research has demonstrated that geniposide enhances the expression of NRF2 and HMOX1, reducing cadmium-induced oxidative stress damage in osteoblasts (He et al., 2019). The NF- κ B/HIF-1 α signaling pathway mediated by HMOX1 exerts anti-inflammatory effects (Jin et al., 2021). Upregulating HMOX1 through the PI3-kinase-JNK-1/2-Nrf2 pathway can enhance the anti-inflammatory capacity of macrophages (Jeon et al., 2011). This is similar to our findings. In summary, geniposide may regulate bone metabolism by inhibiting inflammation, with targeting HMOX1 being one of its key mechanisms of action.

Our study presents several limitations that warrant consideration.

Firstly, this study used multi-omics analysis to explore the key role of HMOX1 in cadmium exposure. These results suggest correlational, rather than definitive causal, regulatory relationships. Inflammatory responses, M2 macrophage polarization, and other factors exhibit complex interactions in cadmium-induced OP, and their biochemical dynamics require further experimental and systems biology exploration. Secondly, the population in the MR study was focused on the European population, so extrapolating these findings to populations with other genetic backgrounds, such as Asians, should be done with caution. Additionally, computational predictions cannot fully capture the complexity of biological systems, and the anti-inflammatory activity of geniposide in OP still requires multi-level validation. These limitations highlight the need for caution in interpreting our findings and underscore the importance of complementary studies to validate and expand upon our conclusions.

In summary, this study explored the potential mechanisms of cadmium-induced OP through multi-omics analysis and identified potential drug targets. The study covered cross-sectional analysis, network toxicology, bioinformatics, single-cell sequencing analysis, and drug-target MR, providing multidimensional support and enhancing the reliability of the results. We found that cadmium exposure is a significant risk factor for osteoporosis, HMOX1 may exert a protective effect by regulating M2 macrophage polarization, and geniposide can target and bind to HMOX1. Our findings not only deepened the understanding of bone loss under cadmium exposure but also provided insights for future drug interventions.

Abbreviation

Osteoporosis OP
National Health and Nutrition Examination Survey NHANES
Gene Set Enrichment Analysis GSEA
Mendelian randomization MR
Swedish cohort of the Osteoporotic Fractures in Men Study MrOS
bone mineral density BMD
Femoral neck bone mineral density FN BMD
Dual-energy X-ray absorptiometry DXA

Comparative Toxicogenomics Database CTD
 Gene Expression Omnibus GEO
 Gene Ontology GO
 biological processes BP
 cellular components CC
 molecular functions MF
 Kyoto Encyclopedia of Genes and Genomes KEGG
 Protein-Protein Interaction PPI
 Support Vector Machine - Recursive Feature Elimination SVM-RFE
 odds ratios OR
 confidence intervals CI
 Area Under the Curve AUC
 decision curve analysis DCA
 Principal component analysis PCA
 Molecular Signatures Database MSigDB
 Genetic Factors for Osteoporosis Consortium GEFOS
 UK Biobank UKB
 Instrumental variables IVs
 single nucleotide polymorphisms SNPs
 transcription start site TSS
 Inverse Variance Weighted IVW
 Protein Data Bank PDB
 Hartree-Fock HF
 canonical ensemble NVT
 isothermal-isobaric ensemble NPT
 root mean square deviation RMSD
 root mean square fluctuation RMSF
 radius of gyration RoG
 molecular mechanics generalized born surface area MM-GBSA
 glutathione GSH
 reactive oxygen species ROS
 small musculoaponeurotic fibrosarcoma sMAF
 iNOS inducible nitric oxide synthase

Authors' contributions

All authors made a significant contribution to the work reported and agreed to be accountable for all aspects of the work. LYW, ZJH and LG designed the experiments. LYW, RYF and JK carried out data extraction LYW, RYF and LXZ carried out mapping and tabulation LYW, JK and LXZ prepared the initial draft of the manuscript LG gave critical feedback during the study or during the submission of the manuscript. All authors provided final approval of the version to be submitted and agreed on the journal for publication.

Clinical trial number

Not applicable.

Data availability

Publicly available datasets and materials were analyzed in this study.

CRediT authorship contribution statement

Yifa Rong: Visualization, Software. **Kai Jiang:** Visualization, Software, Data curation. **Jiahao Zhang:** Data curation, Conceptualization. **Gang Li:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Yiwei Li:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Xuezhen Liang:** Software, Formal analysis, Data curation, Conceptualization.

Consent for publication

All participating authors give their consent for this work to be published.

Ethics approval and consent to participate

All data used in this work are publicly available from studies with relevant participant consent and ethical approval.

Funding

We would like to acknowledge the following financial support: Key Technology Research and Development Program of Shandong Province (NO.2021CXGC010501) and Leading Science and Technology Innovation Team (2024sdscktd-02).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank all the participants involved in this research for their priceless contribution.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2025.118502](https://doi.org/10.1016/j.ecoenv.2025.118502).

Data availability

All data used in this work are publicly available from studies with relevant participant consent and ethical approval. I have shared the link of data.

References

- Adasme, M.F., Linnemann, K.L., Bolz, S.N., Kaiser, F., Salentin, S., Haupt, V.J., Schroeder, M., 2021. PLIP 2021: expanding the scope of the protein-ligand interaction profiler to DNA and RNA. *Nucleic Acids Res.* 49 (W1), W530–W534. <https://doi.org/10.1093/nar/gkab294>.
- Bak, S.U., Kim, S., Hwang, H.J., Yun, J.A., Kim, W.S., Won, M.H., Kim, J.Y., Ha, K.S., Kwon, Y.G., Kim, Y.M., 2017. Heme oxygenase-1 (HO-1)/carbon monoxide (CO) axis suppresses RANKL-induced osteoclastic differentiation by inhibiting redox-sensitive NF- κ B activation. *BMB Rep.* 50 (2), 103–108. <https://doi.org/10.5483/bmbrep.2017.50.2.220>.
- Baran, D.T., Faulkner, K.G., Genant, H.K., Miller, P.D., Pacifici, R., 1997. Diagnosis and management of osteoporosis: guidelines for the utilization of bone densitometry. *Calcif. Tissue Int.* 61 (6), 433–440. <https://doi.org/10.1007/s002239900362>.
- Bowden, J., Davey Smith, G., Burgess, S., 2015. Mendelian randomization with invalid instruments: effect estimation and bias detection through Egger regression. *Int. J. Epidemiol.* 44 (2), 512–525. <https://doi.org/10.1093/ije/dyv080>.
- Bowden, J., Del Greco, M.F., Minelli, C., Zhao, Q., Lawlor, D.A., Sheehan, N.A., Thompson, J., Davey Smith, G., 2019. Improving the accuracy of two-sample summary-data Mendelian randomization: moving beyond the NOME assumption. *Int. J. Epidemiol.* 48 (3), 728–742. <https://doi.org/10.1093/ije/dyy258>.
- Breiman, L., 2001. 2001/10/01. Random Forests. *Mach. Learn.* 45 (1), 5–32. <https://doi.org/10.1023/A:1010933404324>.
- Burgess, S., Bowden, J., Fall, T., Ingelsson, E., Thompson, S.G., 2017. Sensitivity analyses for robust causal inference from mendelian randomization analyses with multiple genetic variants. *Epidemiology* 28 (1), 30–42. <https://doi.org/10.1097/EDE.0000000000000559>.
- Caudill, S.P., Schleicher, R.L., Pirkle, J.L., 2008. Multi-rule quality control for the age-related eye disease study. *Stat. Med.* 27 (20), 4094–4106. <https://doi.org/10.1002/sim.3222>.
- Chang, Y., Jiang, X., Dou, J., Xie, R., Zhao, W., Cao, Y., Gao, J., Yao, F., Wu, D., Mei, H., Zhong, Y., Ge, Y., Xu, H., Jiang, W., Xiao, X., Jiang, Y., Hu, S., Wu, Y., Liu, Y., 2024. Investigating the potential risk of cadmium exposure on seizure severity and anxiety-like behaviors through the ferroptosis pathway in epileptic mice: An integrated multi-omics approach. *J. Hazard Mater.* 480, 135814. <https://doi.org/10.1016/j.jhazmat.2024.135814>.
- Che, J., Yang, J., Zhao, B., Shang, P., 2021. HO-1: A new potential therapeutic target to combat osteoporosis. *Eur. J. Pharm.* 906, 174219. <https://doi.org/10.1016/j.ejphar.2021.174219>.

- Chou, L.C., Tsai, C.C., 2023. Assessing the effectiveness of fermented banana peel extracts for the biosorption and removal of cadmium to mitigate inflammation and oxidative stress. *Foods* 12 (13). <https://doi.org/10.3390/foods12132632>.
- Chudy, P., Bednarczyk, K., Chatian, E., Krzeptowski, W., Szade, K., Żukowska, M., Wolnik, J., Sokołowski, G., Józkowicz, A., & Nowak, W.N. (2024). <https://doi.org/10.1101/2024.11.04.620611>.
- Cortes, C., Vapnik, V.N., 1995. Support-vector networks. *Mach. Learn.* 20, 273–297.
- Davis, S., Meltzer, P.S., 2007. GEOquery: a bridge between the Gene Expression Omnibus (GEO) and BioConductor. *Bioinformatics* 23 (14), 1846–1847. <https://doi.org/10.1093/bioinformatics/btm254>.
- Davis, A.P., Wiegiers, T.C., Johnson, R.J., Sciaky, D., Wiegiers, J., Mattingly, C.J., 2023. Comparative Toxicogenomics Database (CTD): update 2023. *Nucleic Acids Res.* 51 (D1), D1257–D1262. <https://doi.org/10.1093/nar/gkac833>.
- Delano, W.L., 2002. The PyMOL Molecular Graphics System.
- Donertas, H.M., Fabian, D.K., Valenzuela, M.F., Partridge, L., Thornton, J.M., 2021. Common genetic associations between age-related diseases. *Nat. Aging* 1 (4), 400–412. <https://doi.org/10.1038/s43587-021-00051-5>.
- Eddie-Amadi, B.F., Ezejiofor, A.N., Orish, C.N., Rovira, J., Allison, T.A., Orisakwe, O.E., 2022. Banana peel ameliorated hepato-renal damage and exerted anti-inflammatory and anti-apoptotic effects in metal mixture mediated hepatic nephropathy by activation of Nrf2/Hmox-1 and inhibition of Nfkb pathway. *Food Chem. Toxicol.* 170, 113471. <https://doi.org/10.1016/j.fct.2022.113471>.
- Emdin, C.A., Khera, A.V., Kathiresan, S., 2017. Mendelian randomization. *JAMA* 318 (19), 1925–1926. <https://doi.org/10.1001/jama.2017.17219>.
- Freitas, M., Fernandes, E., 2011. Zinc, cadmium and nickel increase the activation of NF-kappaB and the release of cytokines from THP-1 monocytic cells. *Metallomics* 3 (11), 1238–1243. <https://doi.org/10.1039/c1mt00050k>.
- Genchi, G., Sinicropi, M.S., Lauria, G., Carocci, A., Catalano, A., 2020. The effects of cadmium toxicity. *Int. J. Environ. Res. Public Health* 17 (11). <https://doi.org/10.3390/ijerph17113782>.
- Goyal, T., Mitra, P., Singh, P., Sharma, P., Sharma, S., 2021. Evaluation of oxidative stress and pro-inflammatory cytokines in occupationally cadmium exposed workers. *Work* 69 (1), 67–73. <https://doi.org/10.3233/WOR-203302>.
- Greco, M.F., Minelli, C., Sheehan, N.A., Thompson, J.R., 2015. Detecting pleiotropy in Mendelian randomisation studies with summary data and a continuous outcome. *Stat. Med.* 34 (21), 2926–2940. <https://doi.org/10.1002/sim.6522>.
- Guyon, I., Weston, J., Barnhill, S., Vapnik, V., 2002. Gene selection for cancer classification using support vector machines. *Mach. Learn.* 46 (1), 389–422. <https://doi.org/10.1023/A:1012487302797>.
- He, T., Shen, H., Zhu, J., Zhu, Y., He, Y., Li, Z., Lu, H., 2019. Geniposide attenuates cadmium-induced oxidative stress injury via Nrf2 signaling in osteoblasts. *Mol. Med. Rep.* 20 (2), 1499–1508. <https://doi.org/10.3892/mmr.2019.10396>.
- He, N., Zhang, J., Liu, M., Yin, L., 2024. Elucidating the mechanism of plasticizers inducing breast cancer through network toxicology and molecular docking analysis. *Ecotoxicol. Environ. Saf.* 284, 116866. <https://doi.org/10.1016/j.ecoenv.2024.116866>.
- Huang, Y., Li, J., Li, W., Ai, N., Jin, H., 2022. Biliverdin/bilirubin redox pair protects lens epithelial cells against oxidative stress in age-related cataract by regulating NF-kappaB/iNOS and Nrf2/HO-1 pathways. *Oxid. Med. Cell Longev.* 2022, 7299182. <https://doi.org/10.1155/2022/7299182>.
- Jeon, W.K., Hong, H.Y., Kim, B.C., 2011. Genipin up-regulates heme oxygenase-1 via PI3-kinase-JNK1/2-Nrf2 signaling pathway to enhance the anti-inflammatory capacity in RAW264.7 macrophages. *Arch. Biochem. Biophys.* 512 (2), 119–125. <https://doi.org/10.1016/j.abb.2011.05.016>.
- Jia, L., Ma, T., Lv, L., Yu, Y., Zhao, M., Chen, H., Gao, L., 2023. Endoplasmic reticulum stress mediated by ROS participates in cadmium exposure-induced MC3T3-E1 cell apoptosis. *Ecotoxicol. Environ. Saf.* 251, 114517. <https://doi.org/10.1016/j.ecoenv.2023.114517>.
- Jin, S., Guerrero-Juarez, C.F., Zhang, L., Chang, I., Ramos, R., Kuan, C.H., Myung, P., Plikus, M.V., Nie, Q., 2021. Inference and analysis of cell-cell communication using CellChat. *Nat. Commun.* 12 (1), 1088. <https://doi.org/10.1038/s41467-021-21246-9>.
- Jin, C., Lin, B.H., Zheng, G., Tan, K., Liu, G.Y., Yao, Z., Xie, J., Chen, W.K., Chen, L., Xu, T.H., Huang, C.B., Wu, Z.Y., Yang, L., 2022. CORM-3 attenuates oxidative stress-induced bone loss via the Nrf2/HO-1 pathway. *Oxid. Med. Cell Longev.* 2022, 5098358. <https://doi.org/10.1155/2022/5098358>.
- Jin, Z., Zhao, H., Luo, Y., Li, J., Pi, J., He, W., Yan, J., Yang, P., 2021. Geniposide achieved an anti-inflammatory effect through the NF-kB/HIF-1α signaling pathway mediated by VEGFA and HMOX1 genes. *SSRN Electron. J.*
- Jolliffe, I.T., 2002. Principal Component Analysis. Springer. (<https://books.google.com.ar/books?id=TrVF-ao4f88C>).
- Kanis, J.A., McCloskey, E.V., Johansson, H., Oden, A., Melton, L.J., 3rd, Khaltaev, N., 2008. A reference standard for the description of osteoporosis. *Bone* 42 (3), 467–475. <https://doi.org/10.1016/j.bone.2007.11.001>.
- Kim, E.S., Shin, S., Lee, Y.J., Ha, I.H., 2021. Association between blood cadmium levels and the risk of osteopenia and osteoporosis in Korean post-menopausal women. *Arch. Osteoporos.* 16 (1), 22. <https://doi.org/10.1007/s11657-021-00887-9>.
- Kuhn, M., 2008. Building predictive models in R using the caret package. *J. Stat. Softw.* 28, 1–26.
- Kunioka, C.T., Manso, M.C., Carvalho, M., 2022. Association between environmental cadmium exposure and osteoporosis risk in postmenopausal women: a systematic review and meta-analysis. *Int. J. Environ. Res. Public Health* 20 (1). <https://doi.org/10.3390/ijerph20010485>.
- Lei, S.F., Wu, S., Li, L.M., Deng, F.Y., Xiao, S.M., Jiang, C., Chen, Y., Jiang, H., Yang, F., Tan, L.J., Sun, X., Zhu, X.Z., Liu, M.Y., Liu, Y.Z., Chen, X.D., Deng, H.W., 2009. An in vivo genome wide gene expression study of circulating monocytes suggested GBP1, STAT1 and CXCL10 as novel risk genes for the differentiation of peak bone mass. *Bone* 44 (5), 1010–1014. <https://doi.org/10.1016/j.bone.2008.05.016>.
- Li, Y., Huang, B., Ye, T., Wang, Y., Xia, D., Qian, J., 2020. Physiological concentrations of bilirubin control inflammatory response by inhibiting NF-kB and inflammasome activation. *Int. Immunopharmacol.* 84. <https://doi.org/10.1016/j.intimp.2020.106520>.
- Liaw, A., Wiener, M.C., 2007. Classification and Regression by randomForest.
- Liberzon, A., Birger, C., Thorvaldsdottir, H., Ghandi, M., Mesirov, J.P., Tamayo, P., 2015. The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst.* 1 (6), 417–425. <https://doi.org/10.1016/j.cels.2015.12.004>.
- Liu, Z., Wu, J., Dong, Z., Wang, Y., Wang, G., Chen, C., Wang, H., Yang, Y., Sun, Y., Yang, M., Fu, J., Li, J., Zhang, Q., Xu, Y., Pi, J., 2024. Prolonged cadmium exposure and osteoclastogenesis: a mechanistic mouse and in vitro study. *Environ. Health Perspect.* 132 (6), 67009. <https://doi.org/10.1289/EHP13849>.
- Luo, H., Gu, R., Ouyang, H., Wang, L., Shi, S., Ji, Y., Bao, B., Liao, G., Xu, B., 2021. Cadmium exposure induces osteoporosis through cellular senescence, associated with activation of NF-kappaB pathway and mitochondrial dysfunction. *Environ. Pollut.* 290, 118043. <https://doi.org/10.1016/j.envpol.2021.118043>.
- Lv, Z., Zhao, B., Hu, L., Pan, J., Li, X., Li, M., Wu, C., & Mao, W. (2023). <https://doi.org/10.21203/rs.3.rs-3268699/v1>.
- Ma, W., Li, C., 2024. Enhancing postmenopausal osteoporosis: a study of KLF2 transcription factor secretion and PI3K-Akt signaling pathway activation by PIK3CA in bone marrow mesenchymal stem cells. *Arch. Med. Sci.* 20 (3), 918–937. <https://doi.org/10.5114/aoms/171785>.
- Ma, Y., Su, Q., Yue, C., Zou, H., Zhu, J., Zhao, H., Song, R., Liu, Z., 2022. The effect of oxidative stress-induced autophagy by cadmium exposure in kidney, liver, and bone damage, and neurotoxicity. *Int. J. Mol. Sci.* 23 (21). <https://doi.org/10.3390/ijms232113491>.
- McInnes, L., Healy, J., Saul, N., Großberger, L., 2018. UMAP: uniform manifold approximation and projection. *J. Open Source Softw.* 3, 861.
- Naito, Y., Takagi, T., Higashimura, Y., 2014. Heme oxygenase-1 and anti-inflammatory M2 macrophages. *Arch. Biochem. Biophys.* 564, 83–88. <https://doi.org/10.1016/j.abb.2014.09.005>.
- Ougier, E., Fiore, K., Rousselle, C., Assuncao, R., Martins, C., Buekers, J., 2021. Burden of osteoporosis and costs associated with human biomonitoring cadmium exposure in three European countries: France, Spain and Belgium. *Int. J. Hyg. Environ. Health* 234, 113747. <https://doi.org/10.1016/j.ijheh.2021.113747>.
- Ramadan, M.A., Saif Eldin, A.S., 2022. Effect of occupational cadmium exposure on the thyroid gland and associated inflammatory markers among workers of the electroplating industry. *Toxicol. Ind. Health* 38 (4), 210–220. <https://doi.org/10.1177/07482337221085046>.
- Ritchie, M.E., Philipson, B., Wu, D., Hu, Y., Law, C.W., Shi, W., Smyth, G.K., 2015. limma powers differential expression analyses for RNA-seq and microarray studies. *Nucleic Acids Res.* 43 (7), e47. <https://doi.org/10.1093/nar/gkv007>.
- Salomon-Ferrer, R., Götz, A.W., Poole, D., Le Grand, S., Walker, R.C., 2013. Routine microsecond molecular dynamics simulations with AMBER on GPUs. 2. Explicit solvent particle mesh Ewald. *J. Chem. Theory Comput.* 9 (9), 3878–3888. <https://doi.org/10.1021/ct400314y>.
- Satarug, S., Garrett, S.H., Sens, M.A., Sens, D.A., 2010. Cadmium, environmental exposure, and health outcomes. *Environ. Health Perspect.* 118 (2), 182–190. <https://doi.org/10.1289/ehp.0901234>.
- Schwarzer, G., Carpenter, J.R., & Rücker, G. (2015). Meta-Analysis with R. Springer International Publishing. (<https://books.google.com.ar/books?id=xhvFsgEACAAJ>).
- Shannon, P., Markiel, A., Ozier, O., Baliga, N.S., Wang, J.T., Ramage, D., Amin, N., Schwikowski, B., Ideker, T., 2003. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Res.* 13 (11), 2498–2504. <https://doi.org/10.1101/gr.1239303>.
- Slob, E.A.W., Burgess, S., 2020. A comparison of robust Mendelian randomization methods using summary data. *Genet. Epidemiol.* 44 (4), 313–329. <https://doi.org/10.1002/gepi.22295>.
- Su, J., Song, Y., Zhu, Z., Huang, X., Fan, J., Qiao, J., Mao, F., 2024. Cell-cell communication: new insights and clinical implications. *Signal Transduct. Target Ther.* 9 (1), 196. <https://doi.org/10.1038/s41302-024-01888-z>.
- Subramanian, A., Tamayo, P., Mootha, V.K., Mukherjee, S., Ebert, B.L., Gillette, M.A., Paulovich, A., Pomeroy, S.L., Golub, T.R., Lander, E.S., Mesirov, J.P., 2005. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. USA* 102 (43), 15545–15550. <https://doi.org/10.1073/pnas.0506580102>.
- Szklarczyk, D., Kirsch, R., Koutrouli, M., Nastou, K., Mehryary, F., Hachilif, R., Gable, A. L., Fang, T., Doncheva, N.T., Pyysalo, S., Bork, P., Jensen, L.J., von Mering, C., 2023. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res.* 51 (D1), D638–D646. <https://doi.org/10.1093/nar/gkac1000>.
- Tan, S., Su, Y., Huang, L., Deng, S., Yan, G., Yang, X., Chen, R., Xian, Y., Liang, J., Liu, Q., Cheng, J., 2022. Corilagin attenuates osteoclastic osteolysis by enhancing HO-1 and inhibiting ROS. *J. Biochem. Mol. Toxicol.* 36 (7), e23049. <https://doi.org/10.1002/jbt.23049>.
- Tanaka, G., Aminuddin, F., Akhabir, L., He, J.Q., Shumansky, K., Connert, J.E., Anthonisen, N.R., Abboud, R.T., Pare, P.D., Sandford, A.J., 2011. Effect of heme oxygenase-1 polymorphisms on lung function and gene expression. *BMC Med. Genet.* 12, 117. <https://doi.org/10.1186/1471-2350-12-117>.
- Tian, J., Li, Z., Wang, L., Qiu, D., Zhang, X., Xin, X., Cai, Z., Lei, B., 2021. Metabolic signatures for safety assessment of low-level cadmium exposure on human osteoblast-like cells. *Ecotoxicol. Environ. Saf.* 207, 111257. <https://doi.org/10.1016/j.ecoenv.2020.111257>.

- Townsend, B.E., Chen, Y.J., Jeffery, E.H., Johnson, R.W., 2014. Dietary broccoli mildly improves neuroinflammation in aged mice but does not reduce lipopolysaccharide-induced sickness behavior. *Nutr. Res.* 34 (11), 990–999. <https://doi.org/10.1016/j.nutres.2014.10.001>.
- Trott, O., Olson, A.J., 2010. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J. Comput. Chem.* 31 (2), 455–461. <https://doi.org/10.1002/jcc.21334>.
- Verzelloni, P., Urbano, T., Wise, L.A., Vinceti, M., Filippini, T., 2024. Cadmium exposure and cardiovascular disease risk: A systematic review and dose-response meta-analysis. *Environ. Pollut.* 345, 123462. <https://doi.org/10.1016/j.envpol.2024.123462>.
- Vickers, A.J., Elkin, E.B., 2006. Decision curve analysis: a novel method for evaluating prediction models. *Med. Decis. Mak.* 26 (6), 565–574. <https://doi.org/10.1177/0272989X06295361>.
- Vitali, S.H., Fernandez-Gonzalez, A., Nadkarni, J., Kwong, A., Rose, C., Mitsialis, S.A., Kourembanas, S., 2020. Heme oxygenase-1 dampens the macrophage sterile inflammasome response and regulates its components in the hypoxic lung. *Am. J. Physiol. Lung Cell Mol. Physiol.* 318 (1), L125–L134. <https://doi.org/10.1152/ajplung.00074.2019>.
- Vosa, U., Claringbould, A., Westra, H.J., Bonder, M.J., Deelen, P., Zeng, B., Kirsten, H., Saha, A., Kreuzhuber, R., Yazar, S., Brugge, H., Oelen, R., de Vries, D.H., van der Wijst, M.G.P., Kasela, S., Pervjakova, N., Alves, I., Fave, M.J., Agbessi, M., Christiansen, M.W., Jansen, R., Seppala, I., Tong, L., Teumer, A., Schramm, K., Hemani, G., Verloouw, J., Yaghootkar, H., Sonmez Flitman, R., Brown, A., Kukushkina, V., Kalnapenkis, A., Rueger, S., Porcu, E., Kronberg, J., Kettunen, J., Lee, B., Zhang, F., Qi, T., Hernandez, J.A., Arindarto, W., Beutner, F., Consortium, B., i, Dmitrieva, Q.T.L.C., Elansary, J., Fairfax, M., Georges, B.P., Heijmans, M., Hewitt, B.T., Kahonen, A.W., Kim, M., Knight, Y., Kovacs, J.C., Krohn, P., Li, K., Loeffler, S., Mariagorta, M., Mei, U.M., Momozawa, H., Muller-Nurasyid, Y., Nauck, M., Nivard, M., Penninx, M.G., Pritchard, B., Raitakari, J.K., Rotzschke, O.T., Slagboom, O., Stehouwer, E.P., Stumvoll, C.D.A., Sullivan, P., T.M., Hoen, P.A.C., Thiery, J., Tonjes, A., van Dongen, J., van Ijcken, M., Veldink, J.H., Volker, U., Warmerdam, R., Wijmenga, C., Swertz, M., Andiappan, A., Montgomery, G.W., Ripatti, S., Perola, M., Kutalik, Z., Dermitzakis, E., Bergmann, S., Frayling, T., van Meurs, J., Prokisch, H., Ahsan, H., Pierce, B.L., Lehtimäki, T., Boomsma, D.I., Psaty, B.M., Gharib, S.A., Awadalla, P., Milani, L., Ouweland, W.H., Downes, K., Stegle, O., Battle, A., Visscher, P.M., Yang, J., Scholz, M., Powell, J., Gibson, G., Esko, T., Franke, L., 2021. Large-scale cis- and trans-eQTL analyses identify thousands of genetic loci and polygenic scores that regulate blood gene expression. *Nat. Genet.* 53 (9), 1300–1310. <https://doi.org/10.1038/s41588-021-00913-z>.
- Wallin, M., Andersson, E.M., Engstrom, G., 2024. Blood cadmium is associated with increased fracture risk in never-smokers - results from a case-control study using data from the Malmo Diet and Cancer cohort. *Bone* 179, 116989. <https://doi.org/10.1016/j.bone.2023.116989>.
- Wallin, M., Barregard, L., Sallsten, G., Lundh, T., Sundh, D., Lorentzon, M., Ohlsson, C., Mellstrom, D., 2021. Low-level cadmium exposure is associated with decreased cortical thickness, cortical area and trabecular bone volume fraction in elderly men: the MrOS Sweden study. *Bone* 143, 115768. <https://doi.org/10.1016/j.bone.2020.115768>.
- Wang, E., Li, S., Li, Y., Zhou, T., 2025. HMOX1 as a potential drug target for upper and lower airway diseases: insights from multi-omics analysis. *Respir. Res.* 26 (1), 41. <https://doi.org/10.1186/s12931-025-03124-w>.
- Wang, Z., Li, X., Yang, J., Gong, Y., Zhang, H., Qiu, X., Liu, Y., Zhou, C., Chen, Y., Greenbaum, J., Cheng, L., Hu, Y., Xie, J., Yang, X., Li, Y., Schiller, M.R., Chen, Y., Tan, L., Tang, S.Y., Shen, H., Xiao, H.M., Deng, H.W., 2021. Single-cell RNA sequencing deconvolutes the in vivo heterogeneity of human bone marrow-derived mesenchymal stem cells. *Int. J. Biol. Sci.* 17 (15), 4192–4206. <https://doi.org/10.7150/ijbs.61950>.
- Wang, H., Peng, C., Hu, G., Chen, W., Hu, Y., Pi, H., 2024. Integrated single-cell RNA-seq and Bulk RNA-seq identify diagnostic biomarkers for Postmenopausal Osteoporosis. *Curr. Med. Chem.* <https://doi.org/10.2174/0109298673343344240930054414>.
- Watabe, Y., Giam Chuang, V.T., Sakai, H., Ito, C., Enoki, Y., Kohno, M., Otogiri, M., Matsumoto, K., Taguchi, K., 2025. Carbon monoxide alleviates endotoxin-induced acute lung injury via NADPH oxidase inhibition in macrophages and neutrophils. *Biochem. Pharm.* 233, 116782. <https://doi.org/10.1016/j.bcp.2025.116782>.
- Wright, N.C., Looker, A.C., Saag, K.G., Curtis, J.R., Delzell, E.S., Randall, S., Dawson-Hughes, B., 2014. The recent prevalence of osteoporosis and low bone mass in the United States based on bone mineral density at the femoral neck or lumbar spine. *J. Bone Min. Res.* 29 (11), 2520–2526. <https://doi.org/10.1002/jbmr.2269>.
- Xiaofeng, Y., Qinren, C., Jingping, H., Xiao, C., Miaomiao, W., Xiangru, F., Xianxing, X., Meixia, H., Jing, L., Jingyuan, W., Xinxin, C., Hongyu, L., Yanhong, D., Lanxiang, J., Xuming, D., 2012. Geniposide, an iridoid glucoside derived from *Gardenia jasminoides*, protects against lipopolysaccharide-induced acute lung injury in mice. *Planta Med.* 78 (6), 557–564. <https://doi.org/10.1055/s-0031-1298212>.
- Ximenez, J.P.B., Zamarioli, A., Kacena, M.A., Barbosa, R.M., Barbosa, F., Jr, 2021. Association of Urinary and Blood Concentrations of Heavy Metals with Measures of Bone Mineral Density Loss: a Data Mining Approach with the Results from the National Health and Nutrition Examination Survey. *Biol. Trace Elem. Res.* 199 (1), 92–101. <https://doi.org/10.1007/s12011-020-02150-7>.
- Xu, S., Hu, E., Cai, Y., Xie, Z., Luo, X., Zhan, L., Tang, W., Wang, Q., Liu, B., Wang, R., Xie, W., Wu, T., Xie, L., Yu, G., 2024. Using clusterProfiler to characterize multiomics data. *Nat. Protoc.* 19 (11), 3292–3320. <https://doi.org/10.1038/s41596-024-01020-z>.
- Ye, W., Liu, Z., Liu, F., Luo, C., 2021. Heme oxygenase-1 predicts risk stratification and immunotherapy efficacy in lower grade gliomas. *Front. Cell Dev. Biol.* 9, 760800. <https://doi.org/10.3389/fcell.2021.760800>.
- Yeh, P.Y., Li, C.Y., Hsieh, C.W., Yang, Y.C., Wang, P.M., Wung, B.S., 2014. CO-releasing molecules and increased heme oxygenase-1 induce protein S-glutathionylation to modulate NF-kappaB activity in endothelial cells. *Free Radic. Biol. Med.* 70, 1–13. <https://doi.org/10.1016/j.freeradbiomed.2014.01.042>.
- Yin, Y., Weng, Y., Ma, Z., Li, L., 2025. Tectochrysin alleviates periodontitis by modulating M2/M1 macrophage ratio and oxidative stress via nuclear factor Kappa B/heme oxygenase-1/nuclear factor erythroid 2-related factor 2 pathway. *Immunol. Invest.* 54 (1), 97–111. <https://doi.org/10.1080/08820139.2024.2418938>.
- Yoo, M., Shin, J., Kim, J., Ryall, K.A., Lee, K., Lee, S., Jeon, M., Kang, J., Tan, A.C., 2015. DSIgDB: drug signatures database for gene set analysis. *Bioinformatics* 31 (18), 3069–3071. <https://doi.org/10.1093/bioinformatics/btv313>.
- Zabalgoitia, M., Colston, J.T., Reddy, S.V., Holt, J.W., Regan, R.F., Stec, D.E., Rimoldi, J.M., Valente, A.J., Chandrasekar, B., 2008. Carbon monoxide donors or heme oxygenase-1 (HO-1) overexpression blocks interleukin-18-mediated NF-kappaB-PTEN-dependent human cardiac endothelial cell death. *Free Radic. Biol. Med.* 44 (3), 284–298. <https://doi.org/10.1016/j.freeradbiomed.2007.08.012>.
- Zheng, H.F., Forgetta, V., Hsu, Y.H., Estrada, K., Rosello-Diez, A., Leo, P.J., Dahia, C.L., Park-Min, K.H., Tobias, J.H., Kooperberg, C., Kleinman, A., Styrkarsdottir, U., Liu, C.T., Uggla, C., Evans, D.S., Nielson, C.M., Walter, K., Pettersson-Kymmer, U., McCarthy, S., Eriksson, J., Kwan, T., Jhamai, M., Trajanoska, K., Memari, Y., Min, J., Huang, J., Danecek, P., Wilmot, B., Li, R., Chou, W.C., Mokry, L.E., Moayyeri, A., Claussnitzer, M., Cheng, C.H., Cheung, W., Medina-Gomez, C., Ge, B., Chen, S.H., Choi, K., Oei, L., Fraser, J., Kraaij, R., Hibbs, M.A., Gregson, C.L., Paquette, D., Hofman, A., Wibom, C., Tranah, G.J., Marshall, M., Gardiner, B.B., Cremin, K., Auer, P., Hsu, L., Ring, S., Tung, J.Y., Thorleifsson, G., Enneman, A.W., van Schoor, N.M., de Groot, L.C., van der Velde, N., Melin, B., Kemp, J.P., Christiansen, C., Sayers, A., Zhou, Y., Calderari, S., van Rooij, J., Carlson, C., Peters, U., Berlivet, S., Dostie, J., Uitterlinden, A.G., Williams, S.R., Farber, C., Grinberg, D., LaCroix, A.Z., Haessler, J., Chasman, D.I., Giulianini, F., Rose, L.M., Ridker, P.M., Eisman, J.A., Nguyen, T.V., Center, J.R., Nogues, X., Garcia-Giralt, N., Launer, L.L., Gudnason, V., Mellstrom, D., Vandenput, L., Amin, N., van Duijn, C.M., Karlsson, M.K., Ljunggren, O., Svensson, O., Hallmans, G., Rousseau, F., Giroux, S., Bussiere, J., Arp, P.P., Koromani, F., Prince, R.L., Lewis, J.R., Langdahl, B.L., Hermann, A.P., Jensen, J.E., Kaptoge, S., Khaw, K.T., Reeve, J., Formosa, M.M., Xuereb-Anastasi, A., Akesson, K., McGuigan, F.E., Garg, G., Olmos, J.M., Zarrabaita, M.T., Riancho, J.A., Ralston, S.H., Alonso, N., Jiang, X., Goltzman, D., Pastinen, T., Grundberg, E., Gauguier, D., Orwoll, E.S., Karasik, D., Davey-Smith, G., Consortium, A., Smith, A.V., Siggeirsdottir, K., Harris, T.B., Zillikens, M.C., van Meurs, J.B., Thorsteinsdottir, U., Maurant, M.T., Timpon, N.J., Soranzo, N., Durbin, R., Wilson, S.G., Ntzani, E.E., Brown, M.A., Stefansson, K., Hinds, D.A., Spector, T., Cupples, L.A., Ohlsson, C., Greenwood, C.M., Consortium, U.K., Jackson, R.D., Rowe, D.W., Loomis, C.A., Evans, D.M., Ackert-Bicknell, C.L., Joyner, A.L., Duncan, E.L., Kiel, D.P., Rivadeneira, F., Richards, J.B., 2015. Whole-genome sequencing identifies EN1 as a determinant of bone density and fracture. *Nature* 526 (7571), 112–117. <https://doi.org/10.1038/nature14878>.
- Zheng, G.X., Terry, J.M., Belgrader, P., Ryvkin, P., Bent, Z.W., Wilson, R., Ziraldo, S.B., Wheeler, T.D., McDermott, G.P., Zhu, J., Gregory, M.T., Shuga, J., Montesclaros, L., Underwood, J.G., Masquelier, D.A., Nishimura, S.Y., Schnall-Levin, M., Wyatt, P.W., Hindson, C.M., Bharadwaj, R., Wong, A., Ness, K.D., Beppu, L.W., Deeg, H.J., McFarland, C., Loeb, K.R., Valente, W.J., Ericson, N.G., Stevens, E.A., Radich, J.P., Mikkelsen, T.S., Hindson, B.J., Bielas, J.H., 2017. Massively parallel digital transcriptional profiling of single cells. *Nat. Commun.* 8, 14049. <https://doi.org/10.1038/ncomms14049>.
- Zhou, Y., Gao, Y., Xu, C., Shen, H., Tian, Q., Deng, H.W., 2018. A novel approach for correction of crosstalk effects in pathway analysis and its application in osteoporosis research. *Sci. Rep.* 8 (1), 668. <https://doi.org/10.1038/s41598-018-19196-2>.
- Zhou, D., Ran, Y., Yu, R., Liu, G., Ran, D., Liu, Z., 2023. SIRT1 regulates osteoblast senescence through SOD2 acetylation and mitochondrial dysfunction in the progression of Osteoporosis caused by Cadmium exposure. *Chem. Biol. Inter.* 382, 110632. <https://doi.org/10.1016/j.cbi.2023.110632>.
- Zhu, H., Tang, X., Gu, C., Chen, R., Liu, Y., Chu, H., Zhang, Z., 2024. Assessment of human exposure to cadmium and its nephrotoxicity in the Chinese population. *Sci. Total Environ.* 918, 170488. <https://doi.org/10.1016/j.scitotenv.2024.170488>.